1 The table shows population data for a country.

Year	1969	1979	1989	1999	2009
Population in millions (p)	58.81	80.35	105.27	134.79	169.71

The data may be represented by an exponential model of growth. Using t as the number of years after 1960, a suitable model is $p = a \times 10^{kt}$.

(i) Derive an equation for $\log_{10}p$ 10 in terms of a, k and t.

[2]

(ii) Complete the table and draw the graph of $\log_{10}p$ against t, drawing a line of best fit by eye.

[3]

(iii) Use your line of best fit to express $\log_{10}p$ in terms of t and hence find p in terms of t.

[4]

(iv) According to the model, what was the population in 1960?

[1]

(v) According to the model, when will the population reach 200 million?

[3]

2

(i) Sketch the graph of $y = 3^x$.

[2]

(ii) Solve the equation $3^{5x-1} = 500000$.

[3]

3 A hot drink when first made has a temperature which is 65°C higher than room temperature. The temperature difference, d °C, between the drink and its surroundings decreases by 1.7% each minute.

(i) Show that 3 minutes after the drink is made, $d = 61.7$ to 3 significant figures.

[2]

(ii) Write down an expression for the value of d at time n minutes after the drink is made, where n is an integer.

[1]

(iii) Show that when $d < 3$, n must satisfy the inequality

$$n > \frac{\log_{10}3 - \log_{10}65}{\log_{10}0.983}.$$

Hence find the least integer value of n for which $d < 3$.

[4]

(iv) The temperature difference at any time t minutes after the drink is made can also be expressed as $d = 65 \times 10^{-kt}$, for some constant k. Use the value of d for 1 minute after the drink is made to calculate the value of k. Hence find the temperature difference 25.3 minutes after the drink is made.

[4]

4 The thickness of a glacier has been measured every five years from 1960 to 2010.
The table shows the reduction in thickness from its measurement in 1960.

Year	1965	1970	1975	1980	1985	1990	1995	2000	2005	2010
Number of years since 1960 (t)	5	10	15	20	25	30	35	40	45	50
Reduction in thickness since 1960 (h m)	0.7	1.0	1.7	2.3	3.6	4.7	6.0	8.2	12	15.9

An exponential model may be used for these data, assuming that the relationship between h and t is of the form $h = a \times 10^{bt}$, where a and b are constants to be determined.

(i) Show that this relationship may be expressed in the form $\log_{10}h = mt + c$, stating the values of m and c in terms of a and b.

[2]

(ii) Complete the table of values in the answer book, giving your answers correct to 2 decimal places, and plot the graph of $\log_{10}h$ against t, drawing by eye a line of best fit.

[4]

(iii) Use your graph to find h in terms of t for this model.

[4]

(iv) Calculate by how much the glacier will reduce in thickness between 2010 and 2020, according to the model.

[2]

(v) Give one reason why this model will not be suitable in the long term.

[1]

5 Use logarithms to solve the equation $3^{x+1} = 5^{2x}$. Give your answer correct to 3 decimal places.

[4]

6 Fig. 8 shows the graph of $\log_{10}y$ against $\log_{10}x$ It is a straight line passing through the points (2, 8) and (0,2).

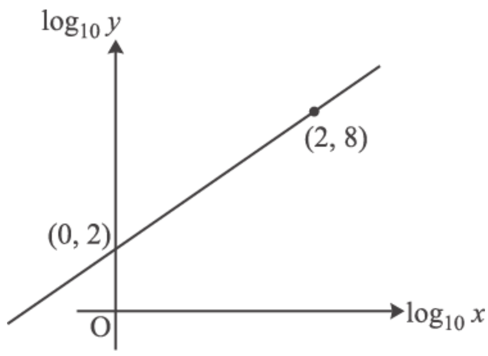


Fig. 8

Find the equation relating $\log_{10}y$ and $\log_{10}x$ and hence find the equation relating y and x.

[4]

7

(i) On the same axes, sketch the curves $y = 3^x$ and $y = 3^{2x}$, identifying clearly which is which.

[3]

(ii) Given that $3^{2x} = 729$, find in either order the values of 3^x and x.

[2]

8 There are many different flu viruses. The numbers of flu viruses detected in the first few weeks of the 2012–2013 flu epidemic in the UK were as follows.

Week	1	2	3	4	5	6	7	8	9	10
Number of flu viruses	7	10	24	32	40	38	63	96	234	480

These data may be modelled by an equation of the form $y = a \times 10^{bt}$, where y is the number of flu viruses detected in week t of the epidemic, and a and b are constants to be determined.

(i) Explain why this model leads to a straight-line graph of $\log_{10}y$ against t. State the gradient and intercept of this graph in terms of a and b.

[3]

(ii) Complete the values of $\log_{10}y$ in the table, draw the graph of $\log_{10}y$ against t, and draw by eye a line of best fit for the data.

Hence determine the values of a and b and the equation for y in terms of t for this model.

[8]

During the decline of the epidemic, an appropriate model was

$$y = 921 \times 10^{-0.137w},$$

where y is the number of flu viruses detected in week w of the decline.

(iii) Use this to find the number of viruses detected in week 4 of the decline.

[1]

9

(i) Simplify $\log_a 1 - \log_a (a^m)^3$.

[2]

(ii) Use logarithms to solve the equation $3^{2x+1} = 1000$. Give your answer correct to 3 significant figures.

[3]

10

Fig. 9 shows the line $y = x$ and the curve $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = \frac{1}{2}(\mathrm{e}^x - 1)$. The line and the curve intersect at the origin and at the point P(a, a).

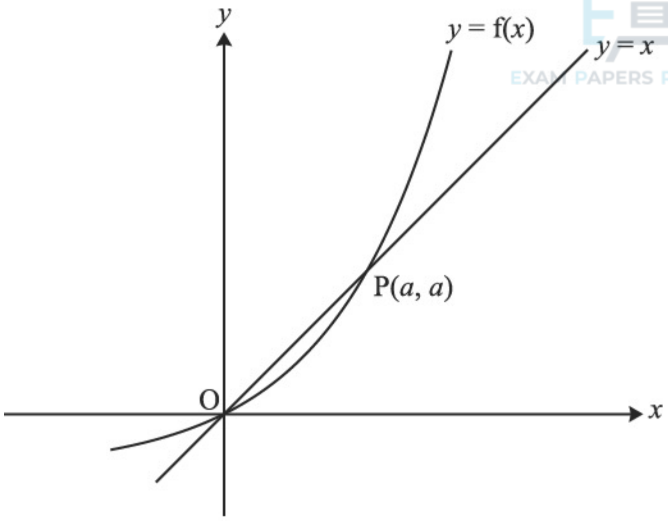


Fig. 9

(i) Show that $e^a = 1 + 2a$.

[1]

(ii) Show that the area of the region enclosed by the curve, the x-axis and the line $x = a$ is $\frac{1}{2}a$. Hence find, in terms of a, the area enclosed by the curve and the line $y = x$.

[6]

(iii) Show that the inverse function of f(x) is g(x), where g(x) = ln(1 + 2x). Add a sketch of y = g(x) to the copy of Fig. 9.

[5]

(iv) Find the derivatives of f(x) and g(x). Hence verify that $g'(a) = \frac{1}{f'(a)}$.

Give a geometrical interpretation of this result.

[7]

11 The temperature θ °C of water in a container after t minutes is modelled by the equation

$$\theta = a - be^{-kt},$$

where a, b and k are positive constants.

The initial and long-term temperatures of the water are 15°C and 100°C respectively. After 1 minute, the temperature is 30°C.

(i) Find a, b and k.

[6]

(ii) Find how long it takes for the temperature to reach 80°C.

[2]

12

Given that $y = \ln\left(\sqrt{\frac{2x-1}{2x+1}}\right)$, show that $\frac{dy}{dx} = \frac{1}{2x-1} - \frac{1}{2x+1}$

[4]

13 Fig. 9 shows the curve $y = xe^{-2x}$ together with the straight line $y = mx$, where m is a constant, with $0 < m < 1$. The curve and the line meet at O and P. The dashed line is the tangent at P.

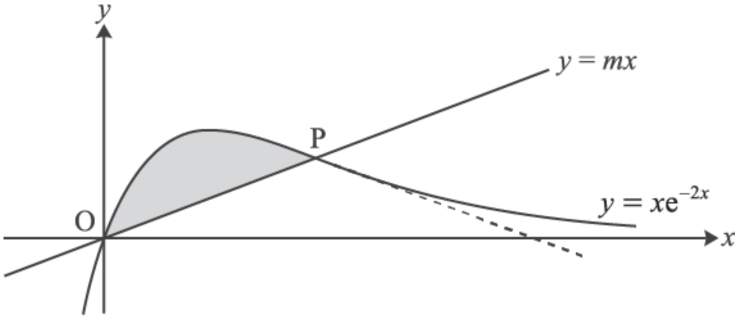


Fig. 9

(i) Show that the x–coordinate of P is $-\frac{1}{2}\ln m$. **[3]**

(ii) Find, in terms of m, the gradient of the tangent to the curve at P. **[4]**

You are given that OP and this tangent are equally inclined to the x-axis.

(iii) Show that $m = e^{-2}$, and find the exact coordinates of P. **[4]**

(iv) Find the exact area of the shaded region between the line OP and the curve. **[7]**

14 The value £V of a car t years after it is new is modelled by the equation $V = Ae^{-kt}$, where A and k are positive constants which depend on the make and model of the car.

(i) Brian buys a new sports car. Its value is modelled by the equation

$$V = 20000\,e^{-0.2t}.$$

Calculate how much value, to the nearest £100, this car has lost after 1 year.

[2]

(ii) At the same time as Brian buys his car, Kate buys a new hatchback for £15 000. Her car loses £2000 of its value in the first year. Show that, for Kate's car, $k = 0.143$ correct to 3 significant figures.

[3]

(iii) Find how long it is before Brian's and Kate's cars have the same value.

[3]

15 Fig. 9 shows the curve $y = f(x)$, where

$$f(x) = (e^x - 2)^2 - 1,\ \ x \in \mathbb{R}.$$

The curve crosses the x-axis at O and P, and has a turning point at Q.

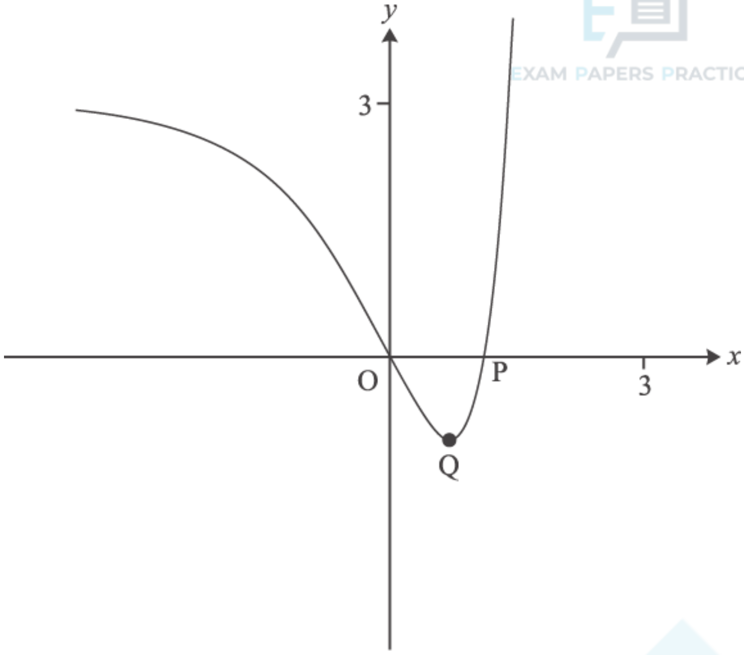


Fig. 9

(i) Find the exact x-coordinate of P.

[2]

(ii) Show that the x-coordinate of Q is ln 2 and find its y-coordinate.

[4]

(iii) Find the exact area of the region enclosed by the curve and the x-axis.

[5]

The domain of $f(x)$ is now restricted to $x \geq \ln 2$.

(iv) Find the inverse function $f^{-1}(x)$. Write down its domain and range, and sketch its graph on the copy of Fig. 9.

[7]

16 Fig. 9 shows the curve $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = \mathrm{e}^{2x} + k\,\mathrm{e}^{-2x}$ and k is a constant greater than 1.

The curve crosses the y-axis at P and has a turning point Q.

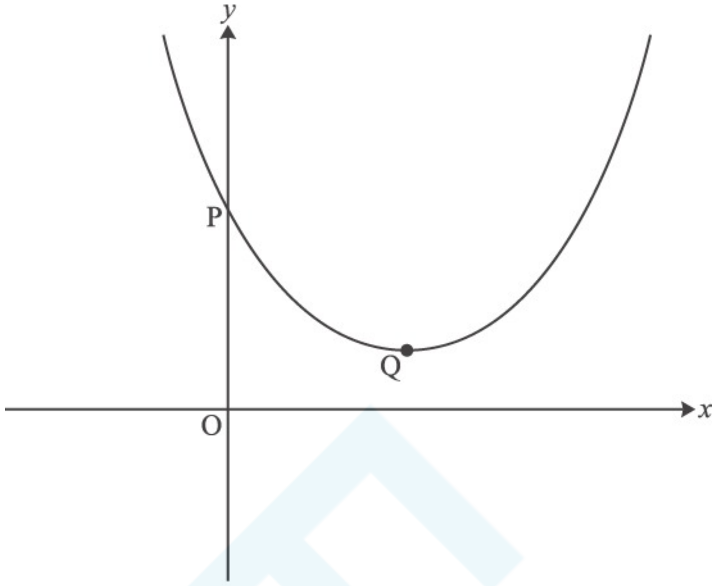


Fig. 9

(i) Find the y-coordinate of P in terms of k.

[1]

(ii) Show that the x-coordinate of Q is $\frac{1}{4}\ln k$, and find the y-coordinate in its simplest form.

[5]

(iii) Find, in terms of k, the area of the region enclosed by the curve, the x-axis, the y-axis and the line $x = \frac{1}{2}\ln k$. Give your answer in the form $ak + b$.

[4]

The function $\mathrm{g}(x)$ is defined by $\mathrm{g}(x) = \mathrm{f}(x + \frac{1}{4}\ln k)$.

(iv)

A Show that $\mathrm{g}(x) = \sqrt{k}\,(\mathrm{e}^{2x} + \mathrm{e}^{-2x})$.

[3]

B Hence show that $\mathrm{g}(x)$ is an even function.

[2]

C Deduce, with reasons, a geometrical property of the curve $y = f(x)$.

[3]

17 The functions $f(x)$ and $g(x)$ are defined by $f(x) = \ln x$ and $g(x) = 2 + e^x$, for $x > 0$.

Find the exact value of x, given that $fg(x) = 2x$.

[5]

18 A biologist is investigating the growth of bacteria in a piece of bread. He believes that the number, N, of bacteria after t hours may be modelled by the relationship $N = A \times 2^{kt}$, where A and k are constants.

(a) Show that, according to the model, the graph of $\log_{10} N$ against t is a straight line.
Give, in terms of A and k,
- the gradient of the line
- the intercept on the vertical axis. **[4]**

The biologist measures the number of bacteria at regular intervals over 22 hours and plots a graph of $\log_{10} N$ against t. He finds that the graph is approximately a straight line with gradient 0.20; the line crosses the vertical axis at 2.0.

(b) Find the values of A and k. **[2]**

(c) Use the model to predict the number of bacteria after 24 hours. **[1]**

(d) Give a reason why the model may not be appropriate for large values of t. **[1]**

19

Evaluate $\int_0^1 \frac{1}{1+\sqrt{x}}\,\mathrm{d}x$, giving your answer in the form $a + b \ln c$, where a, b and c are integers. **[6]**

20 (a) Express $2\log_3 x + \log_3 a$ as a single logarithm. **[1]**

(b) Given that $2\log_3 x + \log_3 a = 2$, express x in terms of a. **[3]**

21 In an experiment, the temperature of a hot liquid is measured every minute. The difference between the temperature of the hot liquid and room temperature is D °C at time t minutes.

Fig. 8 shows the experimental data.

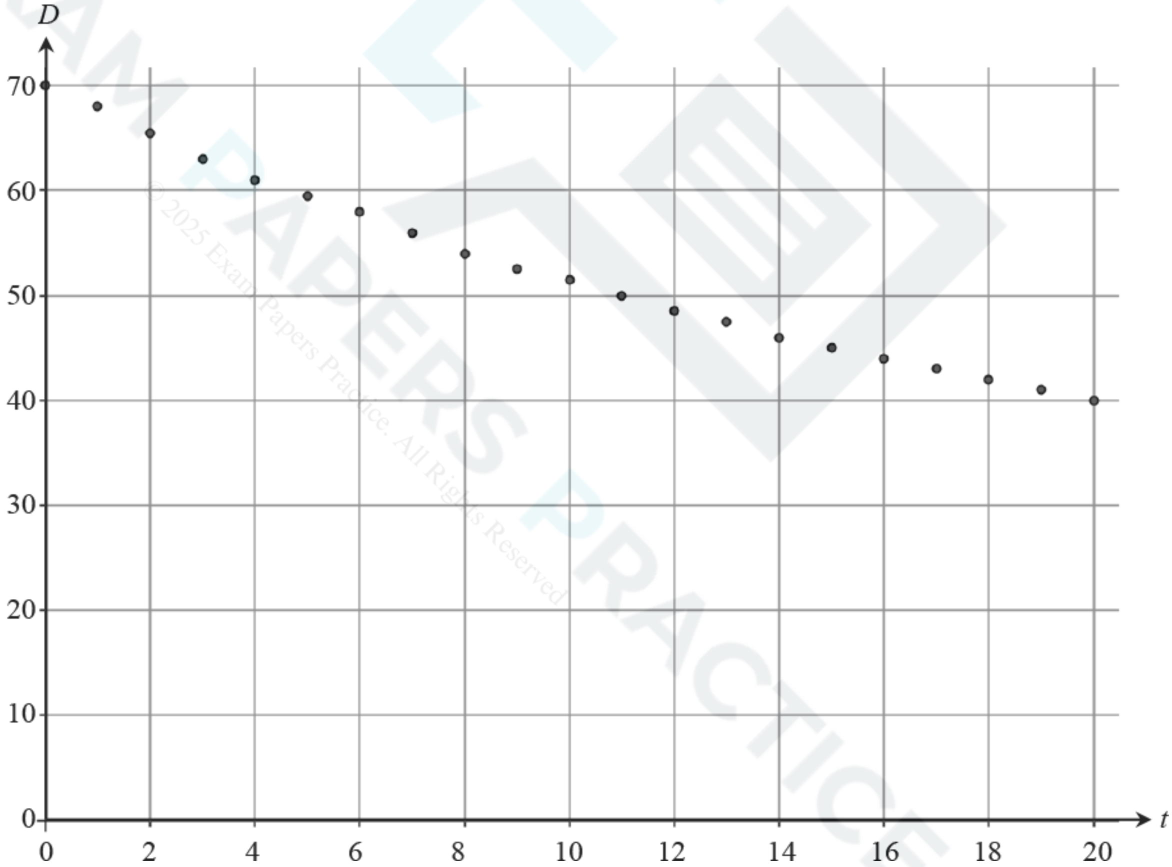


Fig. 8

It is thought that the model $D = 70e^{-0.03t}$ might fit the data.

(a) Write down the derivative of $e^{-0.03t}$. [1]

(b) Explain how you know that $70e^{-0.03t}$ is a decreasing function of t. [1]

(c) Calculate the value of $70e^{-0.03t}$ when

(i) $t = 0$, [1]

(ii) $t = 20$. [1]

(d) Using your answers to parts **(b)** and **(c)**, discuss how well the model $D = 70e^{-0.03t}$ fits the data. [3]

22 In a certain region, the populations of grey squirrels, P_G and red squirrels P_R, at time t years are modelled by the equations:

$$P_G = 10000(1 - e^{-kt})$$

$$P_R = 20000e^{-kt}$$

where $t \geq 0$ and k is a positive constant.

(a) (i) On the axes below, sketch the graphs of P_G and P_R on the same axes.

(ii) Give the equations of any asymptotes. **[4]**

(b) What does the model predict about the long term population of

- grey squirrels
- red squirrels? **[2]**

Grey squirrels and red squirrels compete for food and space. Grey squirrels are larger and more successful than red squirrels.

(c) Comment on the validity of the model given by the equations, giving a reason for your answer. **[1]**

(d) Show that, according to the model, the rate of decrease of the population of red squirrels is always double the rate of increase of the population of grey squirrels. **[4]**

(e) When $t = 3$, the numbers of grey and red squirrels are equal. Find the value of k. **[4]**

23 Show that $\sum_{r=1}^{4} \ln \frac{r}{r+1} = -\ln 5$. **[3]**

24 The curve $y = f(x)$ is defined by the function $f(x) = e^{-x} \sin x$ with domain $0 \le x \le 4\pi$.

(a) (i) Show that the x-coordinates of the stationary points of the curve $y = f(x)$, when arranged in increasing order, form an arithmetic sequence.

(ii) Show that the corresponding y-coordinates form a geometric sequence. **[9]**

(b) Would the result still hold with a larger domain? Give reasons for your answer **[1]**

25 A fisherman has collected statistics for the number of rod-caught salmon in England and Wales. He obtained the following results.

End of year	2004	2005	2006	2007	2008	2009	2015
Number of rod-caught salmon	28 193	21 418	18 776	17 556	16 243	14 526	11 261

Taking y as the number of rod-caught salmon and t as the time in completed years from 2003, the fisherman plotted the graph of $\log_{10}y$ against $\log_{10}t$. This is shown in Fig. 9. The relationship between $\log_{10}y$ and $\log_{10}t$ is modelled by the straight line with equation

$$\log_{10}y = -0.37\log_{10}t + c.$$

This line is also shown in Fig. 9.

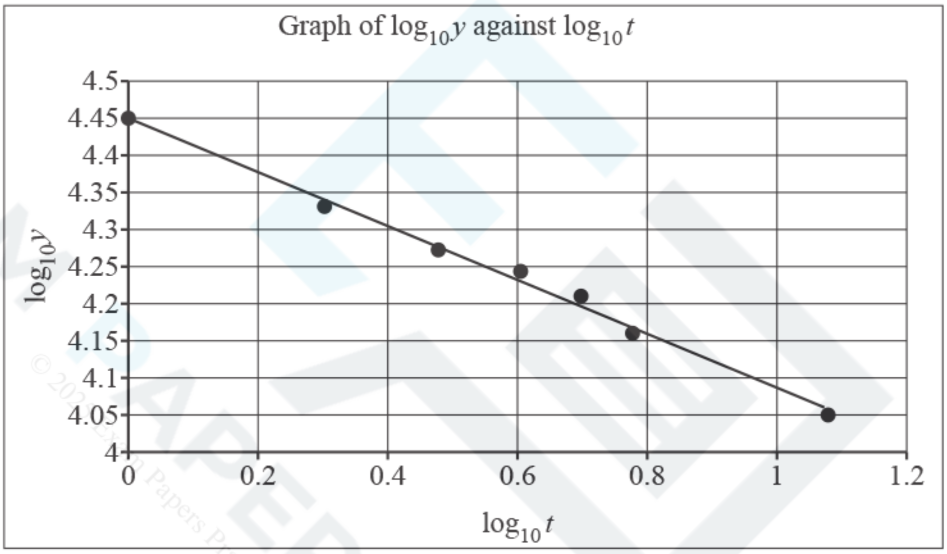


Fig. 9

(a) Use the graph to write down the value of c. **[1]**

(b) Show that $y \approx 28\,200t^{-0.37}$. **[3]**

(c) Verify that the model works well for the year 2006. **[1]**

(d) Use the model to estimate the number of rod-caught salmon in

(i) 2012, **[1]**

(ii) 2025. **[1]**

(e) Comment on the reliability of your answers to part **(d)**. **[2]**

26 **(a)** Sketch the curve $y = e^{2x}$. **[2]**

(b) Describe fully the transformation that maps the curve $y = e^{x}$ onto the curve $y = e^{2x}$. **[2]**

(c) Find the equation of the tangent to $y = e^{2x}$ at the point where $x = 3$, giving your answer in the form $y = e^{a}(bx + c)$ where a, b and c are integers. **[6]**

27 The sequence of positive numbers $a_1, a_2, a_3, \ldots$ is a geometric sequence. Prove that the sequence $\ln a_1, \ln a_2, \ln a_3, \ldots$ is an arithmetic sequence. **[3]**

28 When a particular type of glass is filled to a depth of x cm with champagne, the volume of champagne in the glass is modelled by the formula

$$V = kx^2 - \frac{x^3}{3},$$

where $V = 36$ when $x = 3$.

(a) Find the value of k. **[1]**

When full, a glass contains 140 cm^3 of champagne and the depth is 7.5 cm.

(b) Determine whether the model works well for this value. **[2]**

When Leonie pours champagne into a glass of this type, the volume in the glass after t seconds is modelled by the formula

$$V = 140(1 - e^{-0.2t}).$$

(c) (i) Verify that after 5 seconds the depth of champagne is between 5.20 cm and 5.21 cm. **[2]**

(ii) Find the rate at which the depth of champagne is increasing 5 seconds after she starts pouring. **[4]**

(iii) According to the model, would it be faster for Leonie to pour two half glasses of 70 cm^3 or one full glass of 140 cm^3 of champagne? **[2]**

29 **(i)** Sketch the curve $y = 2^x$ **[2]**

(ii) You are given that $\log_a w = 3 + \log_a x^5 - \log_a 2x + \log_a 6$. Find an expression for w in terms of x and a, giving your answer as simply as possible. **[3]**

30 A liquid is being heated. At time t minutes after heating starts, its temperature, θ °C, is modelled by the equation

$$\theta = 10.5 + 69.5\,(1 - e^{-kt}),$$

where k is a positive constant. The boiling point of the liquid is the value approached by θ as t tends to infinity.

(i) Write down the initial temperature and the boiling point of the liquid. **[2]**

(ii) After being heated for one minute, the liquid has a temperature of 30 °C. Find k. **[3]**

(iii) Find how long it takes from the start of the heating until the temperature is within 1 °C of the boiling point.
Give your answer to the nearest minute.

[3]

31 Initially a car is travelling along a stretch of motorway at a steady speed of 22ms^{-1}. The driver changes lane and accelerates in order to overtake a lorry. The speed of the car, $V\text{ms}^{-1}$, is modelled by the formula

$$V = A + B\,(1 - e^{-0.2t}),$$

where A and B are constants to be determined and t is the time in seconds after the car has started to accelerate.

2 seconds after the car starts to accelerate it is travelling at 25.3ms^{-1}.

(a) Find the values of A and B, giving the value of B correct to 2 significant figures. **[3]**

5 seconds after the car starts to accelerate it is travelling at $28.4 ms^{-1}$.

(b) Determine whether the model is a good fit for the data. **[2]**

(c) According to the model, what is the acceleration of the car at $t = 5$? **[3]**

On this stretch of motorway the speed limit is 110 km h^{-1}. The level of penalty given to a driver caught speeding above 110 km h^{-1} depends on the percentage the speed is above the speed limit. The driver continues to accelerate until reaching the maximum speed predicted by the model.

(d) Determine whether the driver will exceed the speed limit by 10%. **[4]**

32 **In this question you must show detailed reasoning.**

The equation of a curve is $y = \frac{2x}{\ln x}$. Show that the only stationary point on this curve is a minimum point at (e, 2e). **[9]**

33 In a chemical reaction, compound B is formed from compound A and other compounds. The mass of B at time t minutes is x kg. The total mass of A and B is always 1 kg. Sadiq formulates a simple model for the reaction in which the rate at which the mass of B increases is proportional to the product of the masses of A and B.

(a) Show that the model can be written as $\frac{dx}{dt} = kx(1-x)$ where k is a constant. **[1]**

Initially, the mass of B is 0.2 kg.

(b) Solve the differential equation, expressing x in terms of k and t. **[7]**

After 15 minutes, the mass of B is measured to be 0.9 kg.

(c) Find the value of k, correct to 3 significant figures. **[2]**

(d) Find the mass of B after 30 minutes. **[1]**

(e) Explain what the model predicts for the mass of A remaining for large values of t. **[1]**

34 Fig. 3 shows the curve with equation $y = \sqrt{1 + \ln x}$.

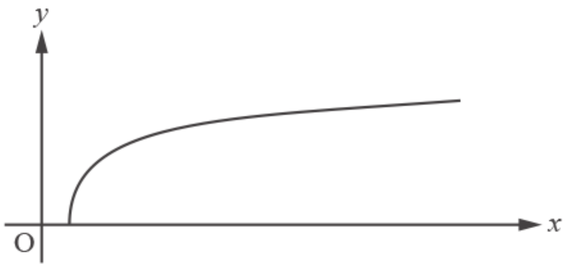


Fig. 3

Determine the exact x-coordinate of the point where the curve meets the x-axis. **[2]**

35 Write down the value of

(a) $\log_a (a^4)$, **[1]**

(b) $\log_a \left(\frac{1}{a}\right)$. **[1]**

36 In an experiment 500 fruit flies were released into a controlled environment. After 10 days there were 650 fruit flies present.

Munirah believes that N, the number of fruit flies present at time t days after the fruit flies are released, will increase at the rate of 4.4% per day. She proposes that the situation is modelled by the formula $N = Ak^{t}$.

(a) Write down the values of A and k. **[2]**

(b) Determine whether the model is consistent with the value of N at $t = 10$. **[2]**

(c) What does the model suggest about the number of fruit flies in the long run? **[1]**

Subsequently it is found that for large values of t the number of fruit flies in the controlled environment oscillates about 750. It is also found that as t increases the oscillations decrease in magnitude.

Munirah proposes a second model in the light of this new information.

$$N = 750 - 250 \times e^{-0.092t}.$$

(d) Identify three ways in which this second model is consistent with the known data. **[3]**

(e) (i) Identify one feature which is not accounted for by the second model. **[1]**

(ii) Give an example of a mathematical function which needs to be incorporated in the model to account for this feature. **[1]**

37 Fig. 10 shows the graph of $y = (k - x)\ln x$ where k is a constant ($k > 1$).

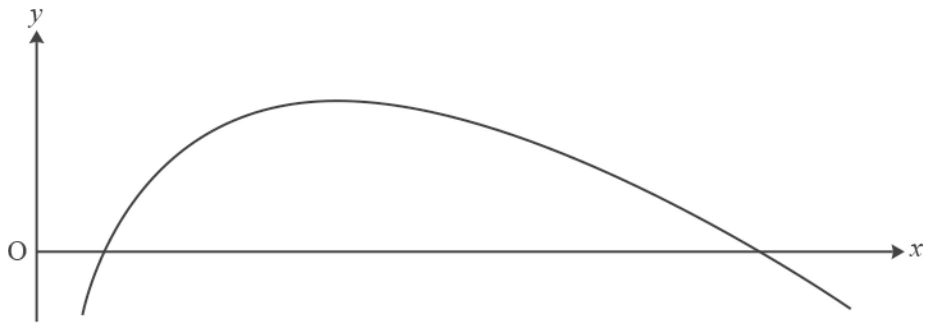


Fig. 10

Find, in terms of k, the area of the finite region between the curve and the x-axis. **[8]**

38 (a) (i) Sketch the graph of $y = 3^x$. [1]

(ii) Give the coordinates of any intercepts. [1]

The curve $y = f(x)$ is the reflection of the curve $y = 3^x$ in the line $y = x$.

(b) Find $f(x)$. [1]

39 In the first year of a course, an A-level student, Aaishah, has a mathematics test each week. The night before each test she revises for t hours. Over the course of the year she realises that her percentage mark for a test, p, may be modelled by the following formula, where A, B and C are constants.

$$p = A - B(t - C)^2$$

- Aaishah finds that, however much she revises, her maximum mark is achieved when she does 2 hours revision. This maximum mark is 62.

- Aaishah had a mark of 22 when she didn't spend any time revising.

(a) Find the values of *A*, *B* and *C*. **[3]**

(b) According to the model, if Aaishah revises for 45 minutes on the night before the test, what mark will she achieve? **[2]**

(c) What is the maximum amount of time that Aaishah could have spent revising for the model to work? **[2]**

In an attempt to improve her marks Aaishah now works through problems for a total of t hours over the three nights before the test. After taking a number of tests, she proposes the following new formula for p.

$$p = 22 + 68(1 - e^{-0.8t})$$

For the next three tests she recorded the data in Fig. 16.

t	1	3	5
p	59	84	89

Fig. 16

(d) Verify that the data is consistent with the new formula. **[2]**

(e) Aaishah's tutor advises her to spend a minimum of twelve hours working through problems in future. Determine whether or not this is good advice. **[2]**

(a) Express $\dfrac{(x^2-8x+9)}{(x+1)(x-2)^2}$ in partial fractions. **[5]**

(b) Express y in terms of x given that

$$\frac{dy}{dx} = \frac{y(x^2 - 8x + 9)}{(x+1)(x-2)^2} \text{ and } y = 16 \text{ when } x = 3.$$

[7]

41 A social media website launched on 1 January 2017. The owners of the website report the number of users the site has at the start of each month. They believe that the relationship between the number of users, n, and the number of months after launch, t, can be modelled by $n = a \times 2^{kt}$ where a and k are constants.

(a) Show that, according to the model, the graph of $\log_{10}n$ against t is a straight line. **[2]**

(b) Fig. 5 shows a plot of the values of t and $\log_{10} n$ for the first seven months. The point at $t = 1$ is for 1 February 2017, and so on.

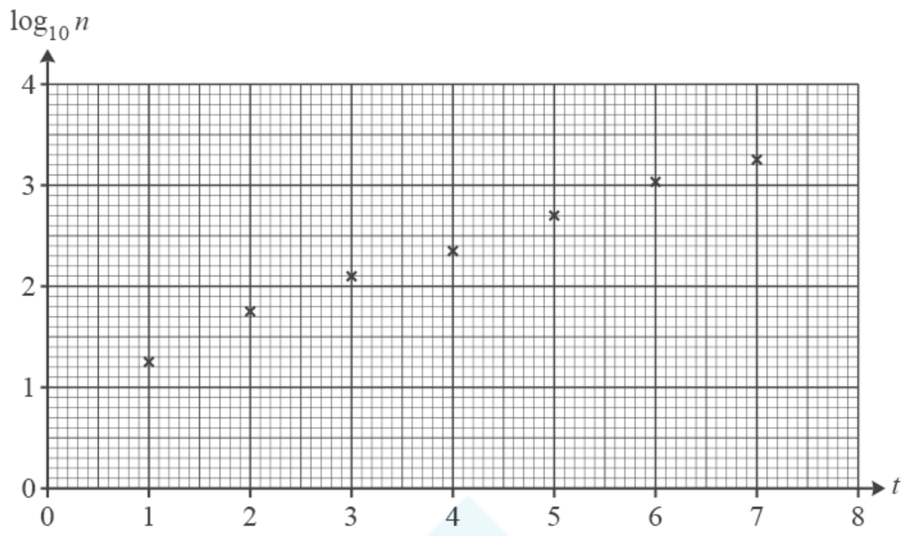


Fig. 5

Find estimates of the values of a and k. **[4]**

(c) The owners of the website wanted to know the date on which they would report that the website had half a million users. Use the model to estimate this date. **[4]**

(d) Give a reason why the model may not be appropriate for large values of t. **[1]**

The function $f(x) = \dfrac{e^x}{1 - e^x}$ is defined on the domain $x \in \mathbb{R}$, $x \neq 0$.

(i) Find $f^{-1}(x)$. **[3]**

(ii) Write down the range of $f^{-1}(x)$. **[1]**

43 The intensity of the sun's radiation, y watts per square metre, and the average distance from the sun, x astronomical units, are shown in Fig. 11 for the planets Mercury and Jupiter.

	x	y
Mercury	0.3075	14400
Jupiter	4.950	55.8

Fig. 11

The intensity y is proportional to a power of the distance x.

(a) Write down an equation for y in terms of x and two constants. **[1]**

(b) Show that the equation can be written in the form $\ln y = a + b \ln x$. **[2]**

(c) Complete the table for ln x and ln y correct to 4 significant figures. **[2]**

	$\ln x$	$\ln y$
Mercury		
Jupiter		

(d) Use the values from part **(c)** to find a and b. **[3]**

(e) Hence rewrite your equation from part **(a)** for y in terms of x, using suitable numerical values for the constants. **[2]**

(f) Sketch a graph of the equation found in part **(e)**. **[2]**

(g) Earth is 1 astronomical unit from the sun. Find the intensity of the sun's radiation for Earth. **[1]**

Fig. 5.1 shows the curve $y = e^{1-x^2}$. Fig. 5.2 shows a spreadsheet used to calculate an estimate of $\int_0^2 e^{1-x^2}\,dx$ using the trapezium rule with four strips.

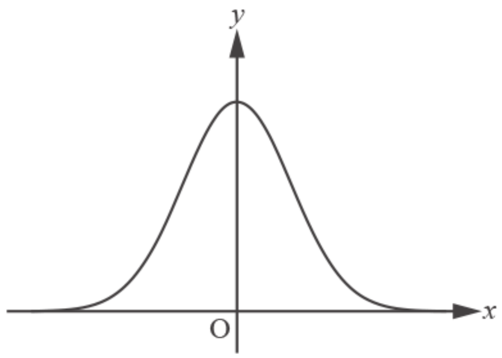


Fig. 5.1

	A	B	C
1	x		y
2	0	1	2.718282
3	0.5	0.75	2.117
4	1	0	1
5	1.5	-1.25	0.286505
6	2	-3	0.049787
7			

Fig. 5.2

(a) Show how the value in cell B3 is calculated. **[1]**

(b) Complete the calculation to estimate $\int_0^2 e^{1-x^2}\,dx$, giving the answer correct to 3 significant figures. **[2]**

(c) Show that the only stationary point on the curve is at (0, e). **[2]**

45 Fig. 7 shows the relationship between the average weekly sales of a newspaper (y) and the time in years after 2010 (x). Each value of y is the average across the whole year. The graph of $\log_{10}y$ plotted against x is approximately a straight line.

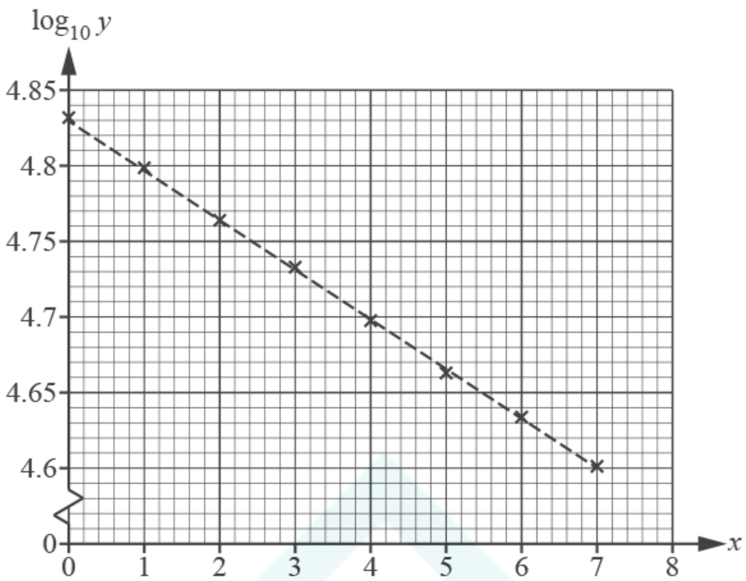


Fig. 7

(a) Show that the straight line is consistent with a model of the form $y = A \times 10^{kx}$, where A and k are constants. **[2]**

(b) Use the straight line to estimate the values of A and k. Giving the answers correct to 3 significant figures. **[4]**

(c) Predict the year in which average weekly sales will fall below 10 000. **[3]**

(d) How reliable do you expect the prediction in part **(c)** to be? Justify your answer. **[1]**

46

Write $\log_a x^5 - \log_a\left(\frac{1}{x}\right)$ in the form $k \log_a x$, where k is a constant to be determined. **[2]**

47

Solve the differential equation $5x\frac{\mathrm{d}y}{\mathrm{d}x} = y^2 - y - 6$ given that $y = 8$ when $x = 1$. Give your answer in the form $y = \mathrm{f}(x)$. **[10]**

48 Chione uses the equation $V = 22000 - a\sqrt{t}$ to model the value of her caravan, where V is the value in pounds t years after purchase.

(a) The model must give the value of the caravan after 4 years as £12 000. Find the value of a. **[1]**

(b) Find the rate at which the value is changing when the caravan is 4 years old. **[3]**

(c) Explain the limitations of this model for large values of t. **[1]**

Chione creates a second model $V = be^{-0.15t} + c$ in which the initial value is £22 000 and the value after a long time tends to £1000.

(d) Find the values of the constants b and c. **[2]**

Chione wishes to find the times other than $t = 0$ at which the two models give the same value.

(e) Show that these times satisfy the equation $t = -\frac{20}{3}\ln\left(1 - \frac{5}{21}\sqrt{t}\right)$. **[3]**

Chione uses fixed point iteration with this equation and $t_0 = 5$. The table shows some of her values.

t_0	5
t_1	
t_2	
t_3	5.157 893 470
t_4	5.187 562 897
t_5	5.210 144 461

(f) Find the missing values t_1 and t_2. [2]

Fig. 14 shows the graphs of $y = t$ and $y = -\frac{20}{3}\ln\left(1 - \frac{5}{21}\sqrt{t}\right)$.

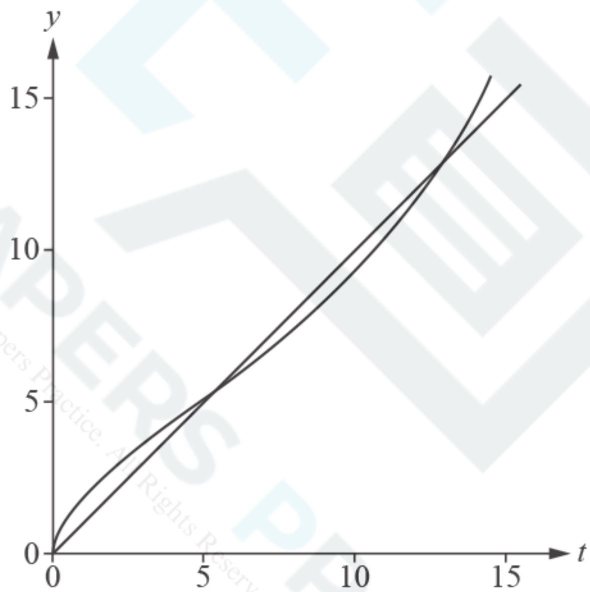
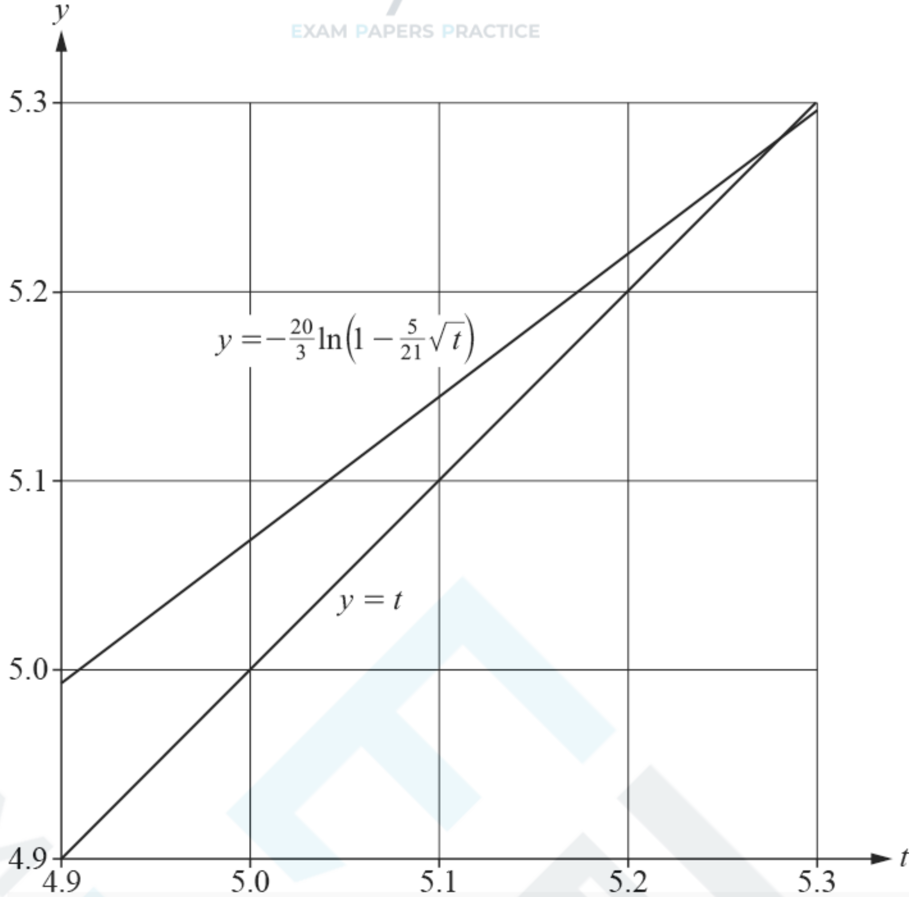


Fig. 14

(g) Sketch the staircase diagram for Chione's iteration on the diagram below. **[1]**



(h) Chione notices that t_4 and t_5 both round to 5.2 so argues that the root of the equation is 5.2 correct to 1 decimal place. Comment on the validity of her argument. **[1]**

(i) Determine whether this fixed point iteration can be used to find the root near to 12. **[2]**

49 **In this question you must show detailed reasoning.**

The curve $y = \ln x$ passes through the point (a, b), where $a > 1$.

The area A is bounded by the x-axis, the line $x = a$ and the curve $y = \ln x$.

The area B is bounded by the x-axis, the y-axis, the line $y = b$ and the curve $y = \ln x$.

The area A is equal in magnitude to the area B.

(a) Show that a satisfies the equation $pa \ln a + qa + r = 0$, where p, q and r are constants to be determined. **[7]**

The value of a is found using the Newton-Raphson method on a spreadsheet. The output is shown in Fig. 15.

r	x_r
0	4
1	5.177399
2	4.931531
3	4.921571
4	4.921554

Fig. 15

Heidi states that the value of a is 4.921554 correct to 6 decimal places.

(b) Determine whether she is correct. [2]

50 The population of a small country is modelled using the formula $P = 5 \times 1.02^n$ where P is the population in millions and n is the number of years after the start of the year 2000.

(a) According to the model, what is the population of the country at the start of the year 2000? **[1]**

(b) Explain fully what the model implies about how the population changes over time. **[2]**

(c) **In this question you must show detailed reasoning.**

According to the model, in what year will the population reach 10 million? **[3]**

(d) Show that, according to the model, the graph of $\log_{10}P$ against n will be a straight line. **[2]**

51 **In this question you must show detailed reasoning.**

(a) Find the exact coordinates of the stationary point of the curve $y = x \ln x$. **[5]**

(b) Show that the stationary point is a minimum turning point. **[2]**

52 (See Insert for H640/03, Practice 4.)

Show that the fractal dimension of the dragon curve is 2, as stated in line 58. **[3]**

53(a) In this question you must show detailed reasoning.

Express $\ln 3 \times \ln 9 \times \ln 27$ in terms of $\ln 3$. **[2]**

(b) Hence show that $\ln 3 \times \ln 9 \times \ln 27 > 6$. **[2]**

54(a) In 2012 Adam bought a second hand car for £8500. Each year Adam has his car valued. He believes that there is a non-linear relationship between t, the time in years since he bought the car, and V, the value of the car in pounds. Fig. 9.1 shows successive values of V and $\log_{10} V$.

t	0	1	2	3	4
V	8500	6970	5720	4690	3840
$\log_{10} V$	3.93	3.84	3.76	3.67	3.58

Fig. 9.1

Adam uses a spreadsheet to plot the points (t, $\log_{10} V$) shown in Fig. 9.1, and then generates a line of best fit for these points. The line passes through the points (0, 3.93) and (4, 3.58). A copy of his graph is shown in Fig. 9.2.

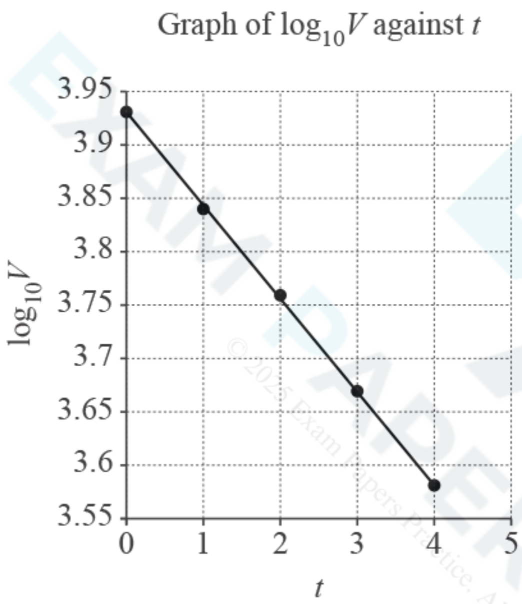


Fig. 9.2

Find an expression for $\log_{10} V$ in terms of t. [3]

(b) Find a model for V in the form $V = A \times b^t$, where A and b are constants to be determined. Give the values of A and b correct to 2 significant figures. **[3]**

(c) In 2017 Adam's car was valued at £3150.

Determine whether the model is a good fit for this data. **[1]**

(d) A company called Webuyoldcars pays £500 for any second hand car. Adam decides that he will sell his car to this company when the annual valuation of his car is less than £500.

According to the model, after how many years will Adam sell his car to Webuyoldcars? **[3]**

55(a) Fig. 8.1 shows the cross-section of a straight driveway 4 m wide made from tarmac.

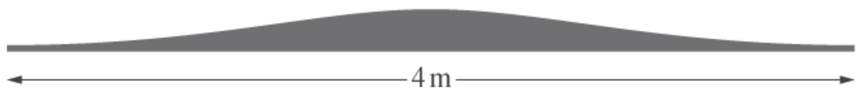


Fig. 8.1

The height h m of the cross-section at a displacement x m from the middle is modelled by $h = \frac{0.2}{1+x^2}$ for $-2 \leqslant x \leqslant 2$.

A lower bound of 0.3615 m^2 is found for the area of the cross-section using rectangles as shown in Fig. 8.2.

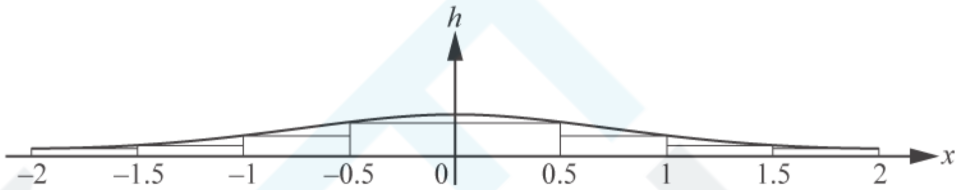


Fig. 8.2

Use a similar method to find an upper bound for the area of the cross-section. [3]

(b)

Use the trapezium rule with 4 strips to estimate $\int_0^2 \frac{0.2}{1+x^2}\,dx$. **[2]**

(c) The driveway is 10 m long. Use your answer in part **(b)** to find an estimate of the volume of tarmac needed to make the driveway. **[2]**

56(a) **In this question you must show detailed reasoning.**

Solve the inequality $\left(\frac{1}{2}\right)^x < 4$. **[3]**

(b) Show that the equation $2\log_2(x+8) - \log_2(x+6) = 3$ has only one root. **[5]**

57(a) **(i)** Write down the derivative of e^{kx}, where k is a constant. **[1]**

(ii) A business has been running since 2009. They sell maths revision resources online.
Give a reason why an exponential growth model might be suitable for the annual profits for the business. **[1]**

(b) Fig. 6 shows the relationship between the annual profits of the business in thousands of pounds (y) and the time in years after 2009 (x). The graph of $\ln y$ plotted against x is approximately a straight line.

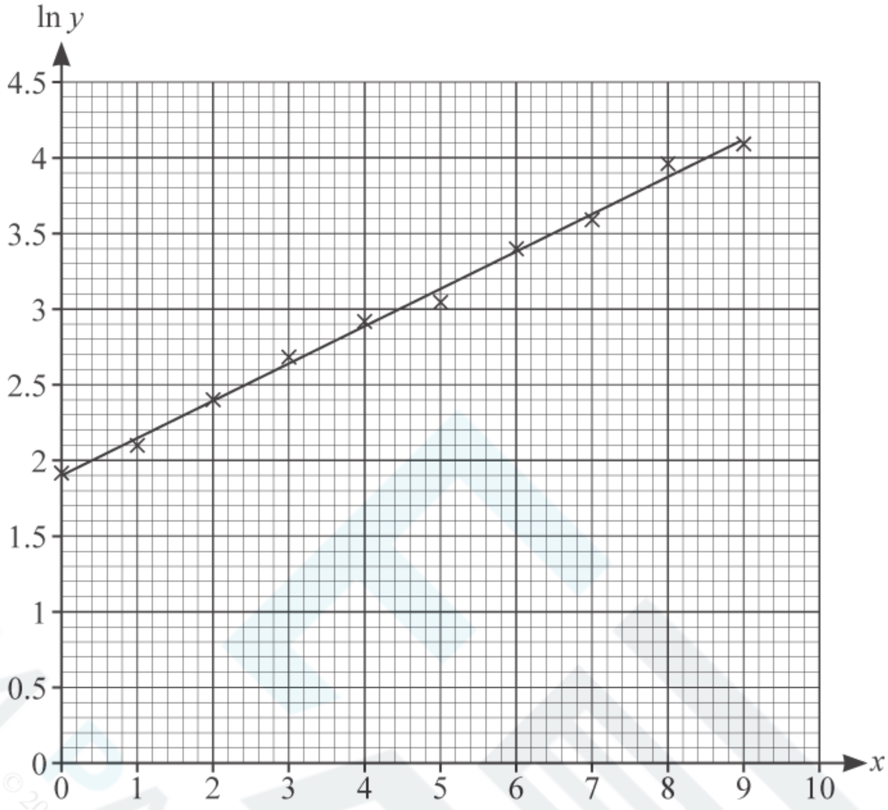


Fig. 6

Show that the straight line is consistent with a model of the form $y = Ae^{kx}$, where A and k are constants. **[2]**

(c) Estimate the values of A and k. **[4]**

(d) Use the model to predict the profit in the year 2020. **[3]**

(e) How reliable do you expect the prediction in part **(d)** to be? Justify your answer. **[1]**

58(a) Please refer to Insert H640-03 Maths B November 2020 to answer this question.

Show that the only stationary point on the curve $y = \frac{\ln x}{x}$ occurs where x = e, as given in line 45. **[3]**

(b) Show that the stationary point is a maximum. **[3]**

(c)

It follows from part **(b)** that, for any positive number a with $a \neq \mathrm{e}$, $\dfrac{\ln \mathrm{e}}{\mathrm{e}} > \dfrac{\ln a}{a}$.

Use this fact to show that $\mathrm{e}^{a} > a^{\mathrm{e}}$. **[2]**

59(a)

Express $\frac{1}{x}+\frac{1}{A-x}$ as a single fraction. [1]

(b) The population of fish in a lake is modelled by the differential equation

$$\frac{dx}{dt} = \frac{x(400-x)}{400}$$

where x is the number of fish and t is the time in years.

When $t = 0$, $x = 100$.

In this question you must show detailed reasoning.

Find the number of fish in the lake when $t = 10$, as predicted by the model. **[8]**

60(a) A car is travelling along a stretch of road at a steady speed of 11ms^{-1}.

The driver accelerates, and t seconds after starting to accelerate the speed of the car, V, is modelled by the formula

$V = A + B(1 - e^{-0.17t})$.

When $t = 3$, $V = 13.8$.

Find the values of A and B, giving your answers correct to 2 significant figures. **[3]**

(b) When $t = 4$, $V = 14.5$ and when $t = 5$, $V = 14.9$.

Determine whether the model is a good fit for these data. **[2]**

(c) Determine the acceleration of the car according to the model when $t = 5$, giving your answer correct to 3 decimal places. **[2]**

(d) The car continues to accelerate until it reaches its maximum speed.

The speed limit on this road is $60 \mathrm{kmh}^{-1}$. All drivers who exceed this speed limit are recorded by a speed camera and automatically fined £100.

Determine whether, according to the model, the driver of this car is fined £100. **[3]**

61 Fig. 5 shows part of the curve $y = \operatorname{cosec} x$ together with the x- and y-axes.

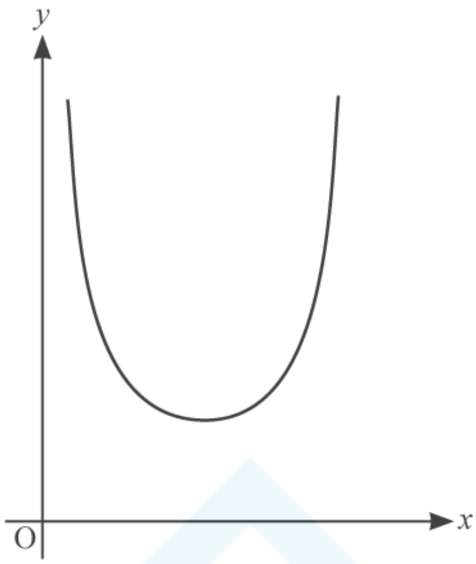


Fig. 5

Show that the equation $x = \operatorname{cosec} x$ has a root which lies between $x = 1$ and $x = 2$. **[2]**

62(a) **In this question you must show detailed reasoning.**

The equation of a curve is

$$y = \frac{\sin 2x - x}{x \sin x}.$$

Use the small angle approximation given below and in the Formula Booklet:

$\sin\theta \approx \theta$, $\cos\theta \approx 1 - \frac{1}{2}\theta^2$, $\tan\theta \approx \theta$ where θ is measured in radians to show that

$$\int_{0.01}^{0.05} y\,dx \approx \ln 5.$$

[4]

(b) Use the same small angle approximation to show that

$\frac{\mathrm{d}y}{\mathrm{d}x} \approx -10000$ at the point where x = 0.01. [2]

(c) The equation $y = 0$ has a root near $x = 1$. Joan uses the Newton-Raphson method to find this root. The output from the spreadsheet she uses is shown in Fig. 10.1.

n	0	1	2	3	4	5	6	7
x_n	1	0.958509	0.950084	0.948261	0.94786	0.947772	0.947753	0.947748

Fig. 10.1

Joan carries out some analysis of this output. The results are shown in Fig. 10.2.

x	y
0.9477475	–7.79967E–07
0.9477485	–2.90821E–06
x	y
0.947745	4.54066E–06
0.947755	–1.67417E–05

Fig. 10.2

Consider the information in Fig. 10.1 and Fig. 10.2.

- Write 4.54066E–06 in standard mathematical notation.
- State the value of the root as accurately as you can, justifying your answer. **[3]**

63(a) Fig. 5 shows part of the curve $y = \operatorname{cosec} x$ together with the x- and y-axes.

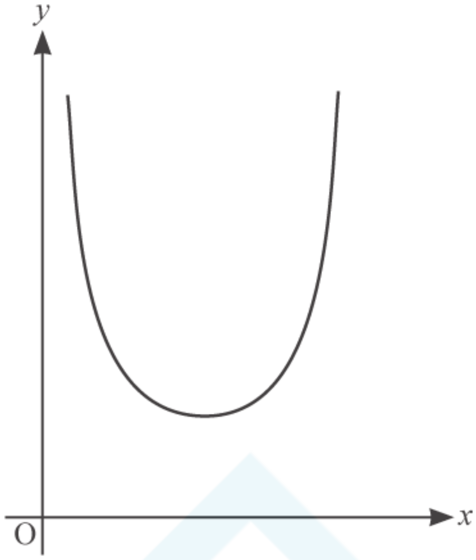
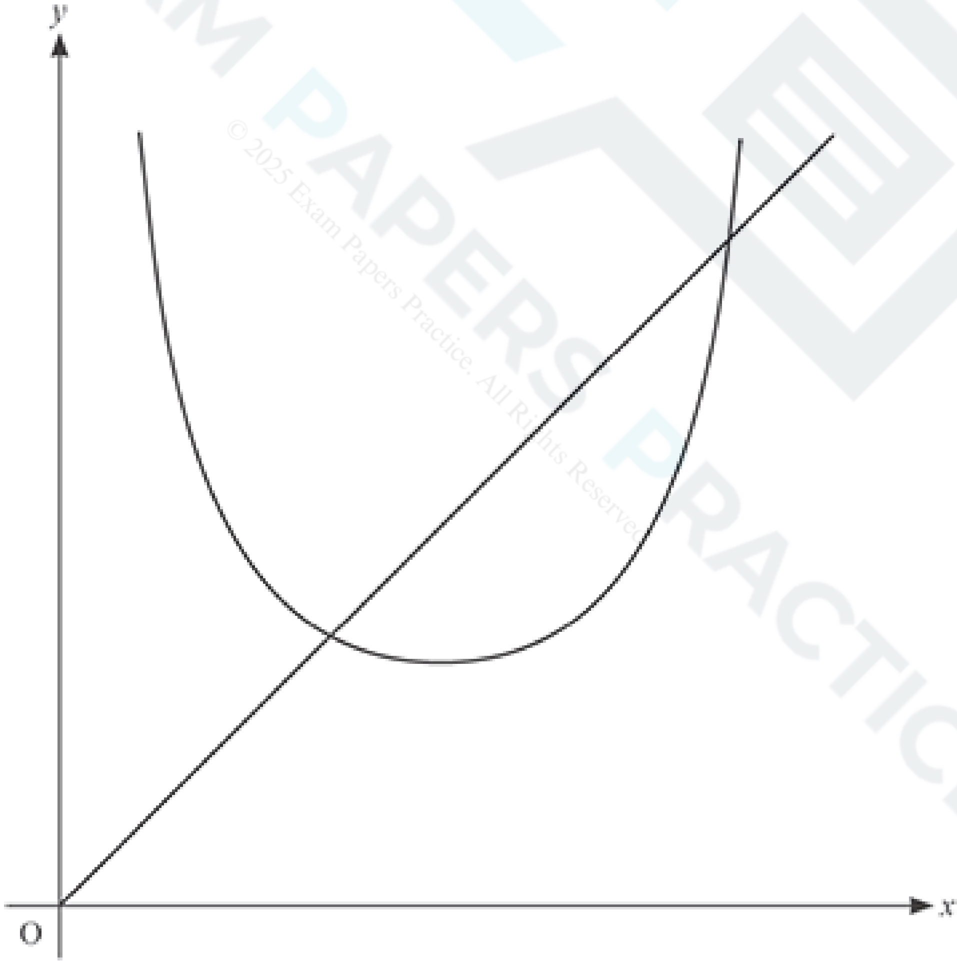


Fig. 5

Sketch the staircase or cobweb diagram for the iteration, starting with $x_0 = 2.5$, on the diagram below. **[3]**



(b) There is another root of $x = \operatorname{cosec} x$ which lies between $x = 2$ and $x = 3$.

Determine whether the iteration $x_{n+1} = \operatorname{cosec}(x_n)$ with $x_0 = 2.5$ converges to this root. **[1]**

(c) Use the iteration $x_{n+1} = \operatorname{cosec}(x_n)$, with $x_0 = 1$, to find

(i) the values of x_1 and x_2, correct to 5 decimal places, **[1]**

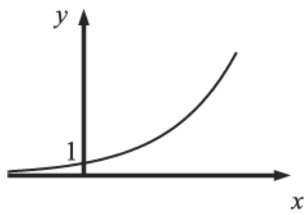
(ii) this root of the equation, correct to 3 decimal places. **[1]**

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Part marks and guidance	
1		i	$\log_{10}p = \log_{10}a + \log_{10}10^{kt}$	M1	condone omission of base;	if unsupported, **B2** for correct equation
		i	$\log_{10}p = \log_{10}a + kt$ www	A1	**Examiner's Comments** The correct equation was often seen, but in many cases it stemmed from wrong working and didn't score. Some candidates stopped at $\log p = \log a + kt\log 10$. $\log p = \log a \times kt$ was a common error; occasionally $\log p = \log a + k\log t$ or logp = loga +logkt were seen.	
		ii	2.02, 2.13, 2.23	B1	allow given to more sig figs	2.022304623 …, 2.129657673, 2.229707433
		ii	plots correct	B1.f.t.	to nearest half square	
		ii	ruled line of best fit	B1	y-intercept between 1.65 and 1.7 and at least one point on or above the line and at least one point on or below the line **Examiner's Comments** This was done very well indeed, with just a few candidates making slips with the plots (usually the middle point), and a few joining each point with a ruler or drawing a curve of best fit to lose the last mark. Only a few candidates lost an easy mark by drawing their line of best fit freehand.	ft their plots must cover range from x = 9 to 49
		iii	0.0105 to 0.0125 for k	B1		must be connected to k
		iii	1.66 to 1.69 for $\log_{10}a$ or 45.7 to 49.0 for a	B1		must be connected to a

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\log_{10}p$ = their kt + their $\log_{10}a$	B1	must be a correct form for equation of line and with their y-intercept and their gradient (may be found from graph or from table, must be correct method)	
		iii	p = their "$47.9 \times 10^{0.0115t}$"or $10^{1.6785 + 0.0115t}$"	B1	as above, "47.9" and "0.0115" must follow from correct method **Examiner's Comments** Most were able to obtain values for the gradient and the y-intercept within the acceptable range, but not all knew what to do with these. For example, log 1.66 or $10^{1.66}$ were often seen in the equation for log$p$. A surprising number of candidates neglected to include an equation for logp at all, and went straight to an equation for p. This was sometimes correct, even if the equation for logp was incorrect. However, a common error was (for example) $p = 45.7 + 10^{0.012t}$.	
		iv	45.7 to 49.0 million	1	'million' needed, not just the value of p **Examiner's Comments** Although many candidates correctly identified the value of loga as crucial in their response, many of them neglected to include the word "million" and lost an easy mark.	
		v	reading from graph at 2.301..	M1*	or $\log_{10}200$ = "$\log_{10}a + kt$"	or 200 = "$10^{\log a + kt}$" oe

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	their 54	M1dep*	eg for their $t = \frac{\log 200 - 1.68}{0.0115}$	or **M1** for their $t = \frac{\log \frac{200}{47.9}}{0.0115}$
		v	2014 cao	A1	if unsupported, allow **B3** only if consistent with graph **Examiner's Comments** Most candidates had the sense to revert to their graph. Accurate plotting and a good line of best fit often rewarded them with full marks. However, most candidates used their answer to part (iii) and sometimes lost the final mark due to rounding. A few used 200 000 000 instead of 200 in one of their equations and failed to score.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
2		i		M1	for curve of correct shape in both quadrants	**SC1** for curve correct in 1^{st} quadrant and touching (0,1) or identified in commentary
		i		A1	through (0, 1) shown on graph or in commentary **Examiner's Comments** This was tackled successfully by most. Most sketches were correct in both quadrants, and (0, 1) was often identified. A small number of candidates only sketched the curve in the first quadrant.	
		ii	$5x - 1 = \dfrac{\log_{10} 500\,000}{\log_{10} 3}$	M1	or $5x - 1 = \log_3 500\,000$	condone omission of base 10 use of logs in other bases may earn full marks
		ii	$x = (\dfrac{\log_{10} 500\,000}{\log_{10} 3} + 1) \div 5$	M1	$x = (\log_3 500000 + 1) \div 5$	
		ii	[x =] 2.588 to 2.59	A1	oe; or **B3** www **Examiner's Comments** This was very well done. A correct initial step of log3500 000 or $^{\log 500\,000}/_{\log 3}$ was almost always present. The most common error was to then subtract 1 from each side. Occasionally only 1 term was divided by 5, and again some candidates rounded prematurely and lost the final mark.	if unsupported, **B3** for correct answer to 3 sf or more www
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
3		i	$65 \times (1 - 0.017)^3$ oe	M1	may be longer method finding decrease year by year etc.	NB use of 3×0.017 leads to 61.685, which doesn't score
		i	61.7410… showing more than 3 sf	A1	answer 61.7 given **Examiner's Comments** A surprising number of candidates failed to score any marks. Many of these candidates adopted a 'simple interest' approach and evaluated 65 – 3×0.017×65. A few candidates evaluated 65 – 3×0.017 or wrote $0.017^3 \times 65 = 61.7$. About two thirds of candidates did understand what was required but failed to appreciate the need to show more than 3 significant figures in their answer to 'show that' the value is 61.7 to this precision. $65 \times 0.983^3 = 61.7$ was quite common. A significant minority of candidates adopted a long-winded approach, showing each stage of the change, and were no more successful.	
		ii	[d =] 65×0.983^n oe	B1	e.g. $63.895 \times 0.983^{n-1}$ or $61.7 \times 0.983^{n-3}$ **Examiner's Comments** Fewer than 40% of candidates earned this mark. $65 \times 0.983^{n-1}$ was quite common, but more often than not the response was either non-existent or irrelevant.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$65 \times 0.983^n < 3$ or $\log_{10}(65 \times 0.983^n) < \log_{10}3$ oe	M1*		condone omission of base 10 throughout
		iii	$\log_{10}65 + \log_{10}0.983^n < \log_{10}3$ www	M1dep	may be implied by e.g. $\log_{10}65 + n\log_{10}0.983 < \log_{10}3$	if **M0M0**, **SC1** for $\log_{10}65 + n\log_{10}0.983 < \log_{10}3$ even if < is replaced by e.g. = or > with no prior incorrect log moves
		iii	$[\log_{10}65 + n\log_{10}0.983 < \log_{10}3]$ $n\log_{10}0.983 < \log_{10}3 - \log_{10}65$ and		or $[\log_{10}0.983^n < \log_{10}3 - \log_{10}65]$	NB watch for correct inequality sign at each step
		iii	completion to $n > \dfrac{\log_{10}3 - \log_{10}65}{\log_{10}0.983}$ **AG** www	A1	inequality signs must be correct throughout	reason for change of inequality sign not required

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$n = 180$ cao	B1	**B0** for $n > 180$ Examiner's Comments This was inaccessible to most candidates, at least partly due to lack of success in the first two parts. It was surprising how few took advantage of the mark for obtaining $n = 180$: this request was either ignored, or a decimal answer was presented – although a few wrote $n > 180$. Very few scored all 3 marks for finding the given result. Most who did, had a correct formula from (ii) but had the inequality sign incorrect or used "=". Very few started off correctly, of those who did start correctly, a high proportion lost the third mark for reversing the sign too early. $\log_{10}(65 \times 0.983^n) < \log_{10}3$ very often incorrectly led straight to $\log_{10}(65) \times \log(0.983^n) < \log_{10}3$ which then became $\log_{10}65 + \log_{10}0.983^n < \log_{10}3$. It was pleasing that many of the successful candidates who did score full marks were justifying the reversal of the sign, even though this was not required.	$n > 179.38\ldots$
		iv	$63.895 = 65 \times 10^{-k}$ soi	B1	or $65 \times 0.983 = 65 \times 10^{-k}$	accept 63.895 rot to 3 or 4 sf; **B1** may be awarded for substitution of $t = 1$ after manipulation
		iv	$\log_{10}$(their 63.895) $= \log_{10}65 - k$ or $-k = \log_{10}$(their 0.983)	M1	their 63.895 must be from attempt to reduce 65 by 1.7% at least once	**M1A1A1** may be awarded if other value of t with correct d is used

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	[*k* =] 7.4 × 10^{-3} to 7.45 × 10^{-3}	A1	[*k* =] $-\log_{10}0.983$ isw	
		iv	[*d* =] 42.1… to 42.123 [°C] isw	A1	**Examiner's Comments** This proved more accessible than part (iii). A little under half of candidates were able to correctly substitute the appropriate value for *d* in conjunction with *t* = 1. However, 63.895 = 65 ×10^{-k} leading to log 63.895 = log65 × log10^{-k} was quite common, so the remaining marks were inaccessible. Some candidates went on to earn the method mark, but lost at least one of the accuracy marks due to premature approximation - some candidates lost a mark by omitting to give an explicit statement of the value of *k*. Some lost both A marks because they divided by log65 instead of subtracting. A significant minority omitted the question altogether. In cases where there was an attempt which scored zero, the most common error was to begin with *d* = 1.	NB **B1M1A0A1** is possible; unsupported answers for *k* and / or *d* do not score
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4		i	$\log_{10}h = \log_{10}a + bt$ www	B1		condone omission of base
		i	$m = b$, $c = \log_{10}a$	B1	**Examiner's Comments** Wrong working often spoiled a correct final answer in this question. It was disappointing to see a significant proportion of candidates failing to score both marks on a very standard piece of work.	must be clearly **stated:** linking equations is insufficient
		ii	–0.15, 0[.00], 0.23, 0.36, 0.56, 0.67, 0.78, 0.91, 1.08, 1.2[0]	B2	B1 if 1 error	
		ii	plots correct (tolerance half square)	B1	condone 1 error – see overlay	no ft available for plots
		ii	single ruled line of best fit for values of x from 5 to 50 inclusive	B1	line must not go outside overlay between $x = 5$ and $x = 50$ **Examiner's Comments** This was very well done. A few candidates made errors in the table — usually the first or the penultimate value. A tiny minority gave all values to a different degree of accuracy to the one requested, thus losing two easy marks — although credit was still available for the plots and the line. Most plotted the points adequately and drew a single ruled line of best fit across the whole range of x-values to earn two marks.	
		iii	$-0.3 \leq y$-intercept ≤ -0.22	B1	may be implied by $0.5 \leq a \leq 0.603$	
		iii	valid method to find gradient of line	M1	may be embedded in equation; may be implied by eg m between 0.025 and 0.035	condone values from table; condone slips eg in reading from graph

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	h = their$a$ × $10^{\text{their}bt}$ or $h = 10^{\text{their log } a\ +\ \text{their}bt}$	M1	**Examiner's Comments** Those candidates who used their graph to find the gradient and the intercept often went on to score full marks in this part. Those who adopted other methods such as simultaneous equations often went astray, and obtaining a positive value for the Y-intercept or a large value for the gradient evidently did not cause concern. It would seem that a significant minority did not connect this part with earlier parts of the question..	if B1M1M0, then SC1 for log$h$ = log$a$ + theirbt isw if both values in the acceptable range for A1
		iii	$0.028 \leq b \leq 0.032$ and $0.5 \leq a \leq 0.603$ or $-0.3 \leq \log a \leq -0.22$	A1		
		iv	$a10^{60b} - a10^{50b}$ their values for a and b	M1	or $10^{\log a\ +\ b\ \times\ 60} - 10^{\log a\ +\ b\ \times\ 50}$ or their values for log$a$ and b	condone 15.9 as second term may follow starting with log$h$ = log$a$ + theirbt
		iv	8.0 to 26.1 inclusive	A1	**Examiner's Comments** t = 70, 10 and 55 were all seen, but many candidates used t = 60 successfully with their model, and then subtracted either 15.9 or f(50) to earn both marks. Unfortunately a few candidates stopped at f(60) lost both marks.	NB A0 for estimate without clear valid method using model; both marks available even if a or b or both are outside range in (iii)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	comment on the **continuing reduction** in thickness **and** its consequences	B1	eg in long term, it predicts that reduction in thickness will continue to increase, even when the glacier has completely melted **Examiner's Comments** Many candidates wrote sensible and worthy responses to this question. Unfortunately, many of them failed to score, in spite of their likely truth, as they were vague or missed the point. Candidates were expected to comment on the model continuing to predict an ever increasing rate of reduction in the thickness of the ice, in spite of the fact that at some point all the ice will have melted.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
5			$(x + 1)\log 3 = 2x\log 5$ oe	M1	or $x + 1 = 2x\log_3 5$ or $(x + 1)\log_5 3 = 2x$	allow recovery from omission of brackets in later working
			$\log 3 = x(2\log 5 - \log 3)$ oe	A1	$x(1 - 2\log_3 5) = -1$ oe or $x(2 - \log_5 3) = \log_5 3$ oe	NB $0.477121254 = 0.920818754x - 1.929947041x = -1$ $1.317393806x = 0.682606194..$
			$\dfrac{\log 3}{2\log 5 - \log 3}$ oe	A1	$\dfrac{1}{2\log_3 5 - 1}$ oe or $\dfrac{\log_5 3}{2 - \log_5 3}$ oe	
			0.518 cao	A1	**Examiner's Comments** Most candidates understood the initial step, but many omitted the brackets and never recovered. Many of those who did earn the first mark often made errors in manipulating the equation, and scored no further marks. The best candidates usually went on to score 4/4.	answer only does not score
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6			$m = 3$ seen	B1		
			$\log y = m\log x + 2$ or $\log y = m\log x + \log 100$	M1	or $\log y - 8 = m(\log x - 2)$	condone lack of base; "$c = 2$" is insufficient
			$\log y = \log x^3 + 2$ or $\log y = \log x^3 + \log 100$ or better	M1	or $10^{\log y} = 10^{3\log x + 2}$ or $10^{3\log x + \log 100}$ or better	condone lack of base, but not bases other than 10 unless fully recovered
			$y = 100^{x3}$ or $y = 10^{3\log x + 2}$ or $y = 10^{\log x^3 + 2}$ www isw	A1	$y = 10^{3\log x + \log 100}$ or $y = 10^{\log x^3 + \log 100}$ **Examiner's Comments** A minority of candidates found this question straightforward and produced fully correct solutions. However, the majority struggled or failed to give sufficient detail of their working to earn full credit. A good number found the gradient of the line as 3. Some used $\frac{\log 6}{\log 2}$, indicating the common misconception of the model. $\log y = 3\log x + \log 2$ was very common as a second statement. Those who earned the second mark very often lost the thirdfor statements such as $y = 3x + 2$ (removing all the "logs") or $y = x^3 + 100$, without $\log y = \log x^3 + 2$, or equivalent, having been seen. It is important that eachstep should be shown as correct final answers were often seen following incorrect working, which of course do not score. A few candidates knew that the final model was of the	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					form $y = ax^b$ and also demonstrated that b was the gradient and a was $10^{(\text{the intercept})}$, producing the correct equation relating *y* and *x*. Many of these candidates would have done better to re-read the question as most of them omitted to state the equation relating logy and logx, which was one of the demands of the question.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7		i	both curves with positive gradients in 1st and 2nd quadrants; ignore labels for this mark	M1	do not award if clearly not exponential shape; condone touching negative *x*-axis but not crossing it	consider each curve independently; ignore scales and points apart from (0, 1)
		i	both through (0, 1)	A1		allow if indicated in table of values or commentary if notmarked on graph
		i	$y = 3^{2x}$ above $y = 3^x$ in first quadrant and below it in second	A1	must be clearly labelled, **A0** if wrongly attributed or if coincide for negative *x* from (0, 1) **Examiner's Comments** A small number of candidates drew two curves of totally different shapes, which was surprising, but most knew the correct shape and although many sketches were sloppily presented, and marks were lost through omitting to identify (0, 1) or by allowing the curves to coalesce through the second quadrant.	if **M0** allow **SC1** for one graph fully correct
		ii	$x = 3$	B1	**B0** if wrongly attributed	
		ii	$3^x = 27$	B1	**B0** if wrongly attributed **Examiner's Comments** This was very well done. Nearly all candidates correctly found $x = 3$; a few thenevaluated 3^3 as 6, 9 or 81.	allow $3^3 = 27$ with $x = 3$ stated
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
8		i	$\log_{10}y = \log_{10}a + bt$ www	B1	**B0** for just $\log_{10}y = \log_{10}a + bt\log_{10}10$	allow omission of base throughout question
		i	gradient is b, intercept is $\log_{10}a$ cao	B2	**B1** for one correct; award independently of their equation; must be stated – linking by arrows etc is insufficient; condone $m = b$ and $c = \log a$ **Examiner's Comments** Many scored full marks in this part, but of those who derived the equation, a significant minority did so incorrectly, thus losing the first mark. "bt" was sometimes quoted as the gradient, and "a = intercept" was a common error. Some candidates failed to state the gradient or the intercept, simply drawing lines to their equation or linking with $y = mx + c$. This is insufficient.	ignore t-intercept is $\frac{-\log_{10} a}{b}$ **B0** for gradient is bt
		ii	1.58, 1.8[0], 1.98, 2.37, 2.68	B1	allow values which round to these numbers to 2 dp;	all values must be correct
		ii	all values correct and all plotted accurately	B1	within tolerance on overlay;	
		ii	ruled line of best fit for at least $1 \le t \le 10$	B1	within tolerance on overlay: must not cut red or green line; line between (1, 0.6) and (1, 1.05) at lower limit and between (10, 2.3) and (10, 2.75) at upper limit;	use ruler tool to check if line is ruled where necessary; tolerance: one small square horizontally at each end; not dependent on correct plots

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	evalution of $\frac{\log y_2 - \log y_1}{t_2 - t_1}$ or substitution of (t_1, logy_1) and (t_2, log y_2) in log $y = bt$ + log a to obtain a numerical value for the gradient	M1	(t_1, log y_1) and (t_2, log y_2) are points on their line	condone use of values from table
		ii	$0.14 \leqslant b \leqslant 0.24$	A1	gradient must be identified as b for **A1**	
		ii	$2.5 \leqslant a \leqslant 6.3$	B1	must be identified as a; not from wrong working	
		ii	y = their $a \times 10^{\text{their } b \times t}$ or $y = 10^{\text{their} bt + \text{ their log} a}$ or $10^{\text{their log } a} \times 10^{\text{their} b \times t}$ oe	M1		if **M0A0B0M0** allow **SC3** for substitution directly into given formula to obtain $y = a10^{bt}$ with a **and** b in acceptable range
		ii	a and b or log a and b both in acceptable range	A1	$0.4 \leq \log a \leq 0.8$ **Examiner's Comments** Most completed the table successfully, and went on to plot the points and draw a suitable line of best fit. A few lost an easy first mark through poor calculator skills (2.34 instead of 2.37 was quite common) and some rounded to 1 decimal place. A few candidates drew a curve of best fit, or failed to use a ruler. Most were able to find the gradient of the line for an easy mark, but many failed to link this to b. Similarly, the instruction to find the value of a was often disregarded. Surprisingly, many candidates simply stopped when they had found a and b, thus losing the last two marks.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	260 or 261	B1	**B0** for non-integer answer **Examiner's Comments** The majority of candidates successfully obtained the correct value, but a significant minority lost an easy mark by failing to give the answer in context as an integer.	
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9		i	$\log_a 1 = 0$ soi or $3m\log_a a$ or $\log_a a^{-3m}$ seen	M1	do not condone $3m\log a$	do not allow MR for $(\log_a a^m)^3$
		i	$-3m$ cao	A1	**Examiner's Comments** Most candidates achieved a method mark from $\log_a 1 = 0$, but were often unable to resolve the second term. Surprisingly, a few candidates dealt successfully with $\log_a (a^m)^3$, but not with the first term.	
		ii	$(2x + 1)\log_3 3 = \log_3 1000$ or $2x + 1 = \log_3 1000$ oe	M1	Or $(2x + 1)\log_{10} 3 = \log_{10} 1000$ [= 3]	condone omission of brackets; allow omission of base 10 or consistent use of other base
		ii	$[x =]\dfrac{\log_3 1000 - 1}{2}$ oe	M1	or $[x =]\dfrac{\dfrac{3}{\log_{10} 3} - 1}{2}$ oe	allow one sign error and / or omission of brackets allow recovery from bracket error for **A1**
		ii	2.64 cao; mark the final answer	A1	not from wrong working **Examiner's Comments** This was done very well indeed. A small number of candidates slipped up in making x the subject, and a few lost the final mark by giving the answer correct to three decimal places.	0 if unsupported or for answer obtained by trial and error on $3^{2x+1} = 1000$
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
10		i	At P(a, a) g(a) = a so ½(e^a –1) = a			
		i	⇒ $e^a = 1 + 2a$*	B1	NB AG **Examiner's Comments** This mark was usually earned.	
		ii	$A = \int_0^a \frac{1}{2}(e^x - 1)\,dx$	M1	correct integral and limits	limits can be implied from subsequent work
		ii	$= \frac{1}{2}\left[e^x - x\right]_0^a$	B1	integral of $e^x - 1$ is $e^x - x$	
		ii	= ½ (e^a –a e^0)	A1		
		ii	= ½ (1 + 2a –a –1) = ½ a*	A1	NB AG	
		ii	area of triangle = ½ a^2	B1		
		ii	area between curve and line = ½ a^2–½ a	B1cao	mark final answer **Examiner's Comments** Virtually everyone scored M1 for writing down the correct integral and limits, but many candidates made a meal of trying to integrate ½ (e^x – 1) , with ¼ (e^x – 1)2 not an uncommon wrong answer. Having successfully negotiated this hurdle, using part (i) to derive ½ a was spotted by about 50% of the candidates. Quite a few candidates managed to recover to earn the final 2 marks for ½ (a^2 – a) (without incorrectly simplifying this to½ a!).	
		iii	y = ½(e^x – 1) swap x and y x = ½ (e^y – 1)			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\Rightarrow 2x = e^y - 1$	M1	Attempt to invert — one valid step	merely swapping x and y is not 'one step'
		iii	$\Rightarrow 2x + 1 = e^y$	A1		
		iii	$\Rightarrow \ln(2x + 1) = y$* $\Rightarrow g(x) = \ln(2x + 1)$	A1	$y = \ln(2x + 1)$ or $g(x) = \ln(2x + 1)$ AG	apply a similar scheme if they start with $g(x)$ and invert to get $f(x)$. or g $f(x) = g((e^x - 1)/2)$ M1
		iii	Sketch: recognisable attempt to reflect in			
		iii	$y = x$	M1	through O and (a, a)	$= \ln(1 + e^x - 1) = \ln(e^x)$ A1 $= x$ A1
		iii	Good shape	A1	no obvious inflexion or TP, extends to third quadrant, without gradient becoming too negative **Examiner's Comments** Finding the inverse function proved to be an easy 3 marks for most candidates – candidates are clearly well practiced in this. The graphs were usually recognisable reflections in y = x, but only well drawn examples – without unnecessary maxima or inflections – were awarded the 'A' mark.	similar scheme for fg See appendix for examples
		iv	$f'(x) = ½ e^x$	B1		
		iv	$g'(x) = 2/(2x + 1)$	M1	$1/(2x + 1)$ (or $1/u$ with $u = 2x + 1$) …	
		iv		A1	… × 2 to get $2/(2x + 1)$	
		iv	$g'(a) = 2/(2a + 1)$, $f'(a) = ½ e^a$	B1	either $g'(a)$ or $f'(a)$ correct soi	
		iv	so $g'(a) = 2/e^a$ or $f'(a) = ½ (2a + 1)$	M1	substituting $e^a = 1 + 2a$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	= 1/(½e^a) = (2a + 1)/2 [= 1/f ′(a)] [= 1/g ′(a])	A1	establishing f ′(a) = 1/g ′(a)	either way round
		iv	tangents are reflections in y = x	B1	must mention tangents **Examiner's Comments** This proved to be more difficult, as intended for the final question in the paper. As with the integral, many candidates struggled to differentiate ½ (e^x – 1) correctly, and equally many omitted the '2' in the numerator of the derivative of ln(1 + 2x). Once these were established correctly the substitution of x = a and establishing of f'(a) = 1/g'(a) was generally done well, though sometimes the arguments using the result in part (i) were either inconclusive or done 'backwards'. The final mark proved to be elusive for most, as we needed the word 'tangent' used here to provide a geometric interpretation of the reciprocal gradients.	
			Total	19		

Question			Answer/Indicative content	Marks	Part marks and guidance	
11		i	$\theta = a - be^{-kt}$			
		i	When $t = 0$, $\theta = 15 \Rightarrow 15 = a - b$	M1	$15 = a - b$	must have $e^0 = 1$
		i	When $t = \infty$, $\theta = 100 \Rightarrow 100 = a$	B1	$a = 100$	
		i	$\Rightarrow b = 85$	A1cao	$b = 85$	
		i	When $t = 1$, $\theta = 30 \Rightarrow 30 = 100 - 85e^{-k}$	M1	$30 = a - b\,e^{-k}$	(need not substitute for a and b)
		i	$\Rightarrow e^{-k} = 70/85$			
		i	$\Rightarrow -k = \ln(70/85) = -0.194(156\ldots)$	M1	Re-arranging and taking lns	allow $-k = \ln[(a - 30)/b]$ ft on a, b
		i	$\Rightarrow k = 0.194$	A1	0.19 or better, or $-\ln(70/85)$ oe **Examiner's Comments** In general, this is a well-known topic which is done successfully. Candidates who managed to deduce that a = 100 using $e^{-kt} \to 0$ as $t \to \infty$ usually gained full marks; those who did not often wasted time trying to solve simultaneous equations using $a - b = 15$ and $30 = a - b\,e^{-k}$.	mark final ans
		ii	$80 = 100 - 85\,e^{-0.194t}$ $\Rightarrow e^{-0.194t} = 20/85$	M1	ft their values for a, b and k	but must substitute values

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	⇒ $t = -\ln(4/17) / 0.194 = 7.45$ (min)	A1	art 7.5 or 7 min 30 s or better **Examiner's Comments** There was an easy method mark to be gained from following through their values of a, b and k. Almost all who got these correct in part (i) scored both marks here, though very occasionally a premature rounding of k produced an insufficiently accurate answer.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12			$y=\ln\left(\sqrt{\frac{2x-1}{2x+1}}\right)=\frac{1}{2}(\ln(2x-1)-\ln(2x+1))$	M1	use of ln(a/b) = ln a – ln b□	
				M1	use of ln√c = ½ ln c	
			$\Rightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{2}\left(\frac{2}{2x-1}-\frac{2}{2x+1}\right)$	A1	o.e.; correct expression (if this line of working is missing, M1M1A0A0)	
			$=\frac{1}{2x-1}-\frac{1}{2x+1}$ *	A1	**NB AG** for alternative methods, see additional solutions	
			Additional solutions			
			$y=\ln\left(\sqrt{\frac{2x-1}{2x+1}}\right)=\frac{1}{2}\ln\left(\frac{2x-1}{2x+1}\right)$	M1	ln √c = ½ ln c used	
			$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{2}\frac{1}{\left(\frac{2x-1}{2x+1}\right)}\frac{(2x+1)2-(2x-1)2}{(2x+1)^2}$ $=\frac{1}{2}\frac{2x+1}{2x-1}\frac{4}{(2x+1)^2}=\frac{2}{(2x-1)(2x+1)}$ $\frac{1}{2x-1}-\frac{1}{2x+1}=\frac{2x+1-(2x-1)}{(2x-1)(2x+1)}$ $=\frac{2}{(2x-1)(2x+1)}$	A2	fully correct expression for the derivative	
			$\frac{1}{2x-1}-\frac{1}{2x+1}=\frac{2x+1-(2x-1)}{(2x-1)(2x+1)}=\frac{2}{(2x-1)(2x+1)}$	A1	simplified and shown to be equivalent to $\frac{1}{2x-1}-\frac{1}{2x+1}$	
			Additional solutions			
			$y=\ln\left(\sqrt{\frac{2x-1}{2x+1}}\right)=\ln\sqrt{2x-1}-\ln\sqrt{2x+1}$	M1	ln(a/b) = ln a – ln b used	
			$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{\sqrt{2x-1}}\frac{1}{2}(2x-1)^{-1/2}.2-\frac{1}{\sqrt{2x+1}}\frac{1}{2}(2x+1)^{-1/2}.2$	A2	fully correct expression	
			$=\frac{1}{2x-1}-\frac{1}{2x+1}$	A1	simplified and shown to be equivalent to $\frac{1}{2x-1}-\frac{1}{2x+1}$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Additional solutions $y=\ln\left(\sqrt{\frac{2x-1}{2x+1}}\right)$ $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{\sqrt{\frac{2x-1}{2x+1}}}\frac{1}{2}\left(\frac{2x-1}{2x+1}\right)^{-1/2}\frac{(2x+1)2-(2x-1)2}{(2x+1)^2}$ or	M1	$\frac{1}{u}\times$ their u' where $u=\sqrt{\frac{2x-1}{2x+1}}$ or $\frac{\sqrt{2x-1}}{\sqrt{2x+1}}$	
			$\frac{1}{\frac{\sqrt{2x-1}}{\sqrt{2x+1}}}\frac{\sqrt{2x+1}.\frac{1}{2}.2(2x-1)^{-1/2}-\sqrt{2x-1}.\frac{1}{2}.2(2x+1)^{-1/2}}{\sqrt{2x+1}^2}$ $=\frac{1}{2}\left(\frac{2x+1}{2x-1}\right)\frac{4}{(2x+1)^2}=\frac{2}{(2x-1)(2x+1)}$	A2	(any attempt at u' will do) o.e. any completely correct expression for the derivative or $=\frac{\sqrt{2x+1}}{\sqrt{2x-1}}\frac{(2x+1)-(2x-1)}{(2x+1)^{3/2}(2x-1)^{1/2}}=\frac{2}{(2x+1)(2x-1)}$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$\frac{1}{2x-1}-\frac{1}{2x+1}=\frac{(2x+1)-(2x-1)}{(2x-1)(2x+1)}=\frac{2}{(2x-1)(2x+1)}$	A1	simplified and correctly shown to be equivalent to $\frac{1}{2x-1}-\frac{1}{2x+1}$ Examiner's Comments Some candidates spotted the trick of simplifying the given function to get y = ½ ln(2x – 1) – ½ ln(2x + 1) before differentiating, and thereby made lives considerably easier for themselves! However, writing the answer down from here omitted the vital 2 x ½ working and lost two marks. Those who started differentiating from y = ln(√(2x – 1) – ln(√(2x + 1)) needed to convince that they were using a chain rule on √u, where u = 2x – 1. Some tenacious candidates even managed to differentiate the given function correctly without these preliminaries, but made life hard for themselves.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13		i	$xe^{-2x} = mx$	M1	may be implied from 2nd line	
		i	$\Rightarrow e^{-2x} = m$	M1	dividing by x, or subtracting ln x	o.e. e.g. [ln x]– $2x$ = ln m + [ln x] or factorising: $x(e^{-2x} - m) = 0$
		i	$\Rightarrow -2x = \ln m$			
		i	$\Rightarrow x = -½ \ln m$*	A1	**NB AG**	
		i	or			
		i	If $x = -½ \ln m$, $y = -½ \ln m \times e^{\ln m}$	M1	substituting correctly	
		i	$= -½ \ln m \times m$	A1		
		i	so P lies on $y = mx$	A1	**Examiner's Comments** This was well answered, even by weaker candidates.	
		ii	let $u = x$, $u' = 1$, $v = e^{-2x}$, $v' = -2e^{-2x}$	M1*	product rule consistent with their derivs	
		ii	$dy/dx = e^{-2x} - 2xe^{-2x}$	A1	o.e. correct expression	
		ii	$= e^{-2.(-\frac{1}{2}\ln m)} - 2.(-\frac{1}{2}\ln m)e^{-2.(-\frac{1}{2}\ln m)}$	M1dep	subst $x = -½ \ln m$ into their deriv dep M1*	
		ii	$= e^{\ln m} + e^{\ln m} \ln m$ [$= m + m \ln m$]	A1cao	condone e^{1nm}not simplified **Examiner's Comments** The product rule here was generally well done, followed by substituting x = - ½ ln m, where some sign errors occurred. Some left the eln m terms unresolved, which was condoned here. The main error was to get a derivative of -2xe-2x.	but not – 2(– ½ ln m), but mark final ans
		iii	$m + m1\text{n } m = -m$	M1	their gradient from (ii) = $-m$	
		iii	$\Rightarrow \ln m = -2$			
		iii	$\Rightarrow m = e^{-2}$*	A1	**NB AG**	
		iii	*or*			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$y + ½m\ln m = m(1 + \ln m)(x + ½ \ln m)$ $x = – \ln m$, $y = 0 \Rightarrow ½m\ln m = m(1 + \ln m)(–½ \ln m)$ $\Rightarrow 1 + \ln m = – 1$, $\ln m = – 2$, $m = e^{–2}$	B2	for fully correct methods finding $x$intercept of equation of tangent and equating to $– \ln m$	
		iii	At P, $x = 1$	B1		
		iii	$\Rightarrow y = e^{-2}$	B1	isw approximations **Examiner's Comments** The first two marks here were the least successfully answered, because most candidates were not familiar with the fact that lines equally inclined to the x-axis have gradients m and –m. Only the best candidates found the result successfully. However, many recovered to find the coordinates of P correctly.	not $e^{-2} \times 1$
		iv	Area under curve $= \int_0^1 xe^{-2x}dx$			
		iv	$u = x$, $u' = 1$, $v' = e^{-2x}$, $v = – ½ e^{-2x}$	M1	parts, condone $v = k e^{-2x}$, provided it is used consistently in their parts formula	ignore limits until 3rd A1
		iv	$= \left[-\frac{1}{2}xe^{-2x}\right]_0^1 + \int_0^1 \frac{1}{2}e^{-2x}dx$	A1ft	ft their v	
		iv	$= \left[-\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x}\right]_0^1$	A1	$-\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x}$ o.e	
		iv	$= (– ½ e^{–2} – ¼ e^{–2}) – (0 – ¼ e^0)$ $[= ¼ – ¾ e^{–2}]$	A1	correct expression	need not be simplified

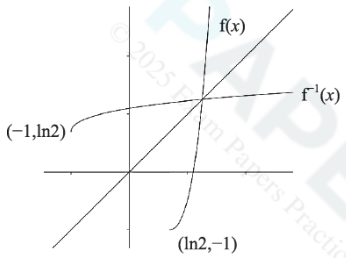
Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	Area of triangle = ½ base × height	M1	ft their 1, e^{-2} or $[e^{-2}x^2/2]$	o.e. using isosceles triangle
		iv	$= ½ × 1 × e^{-2}$	A1		M1 may be implied from 0.067 …
		iv	So area enclosed $= ¼ – 5e^{-2}/4$	A1cao	o.e. must be exact, two terms only **Examiner's Comments** This question tested the more able candidates. The integration by parts required careful control of negative signs and accurate work; the area of the triangle (or integral of the line) were quite often discernable from the working, which was often scrambled and incoherent – perhaps because some candidates were rushing to complete the paper!	isw
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i	$V = 20000e^{-0.2t}$			
		i	when $t = 1$, $V = 16374.615$ …	B1	(soi) art 16400	
		i	so car loses (£)3600	B1	condone no £, must be to nearest £100 **Examiner's Comments** This was a very accessible two marks, provided candidates answered question – the loss in value rounded to the nearest £100.	or B2 for correct answer
		ii	When $t = 1$, $V = 13000$			If $k = 0.143$ verified, e.g.
		ii	$\Rightarrow 13000 = 15000\ e^{-k}$	M1		$15000\ e^{-0.143} = 13001[.31$ …], SCB1
		ii	$\Rightarrow -k\ [\ln e] = \ln(13000/15000)$	M1	taking lns correctly oe e.g. $\ln 13000 = \ln 15000 - k\ [\ln e]$	need not have substituted for V and A
		ii	$\Rightarrow k = 0.1431 \ldots = 0.143$ (3sf) *	A1	cao NB AG must show some working if 4th d.p. not shown **Examiner's Comments** Again, this was very well answered. Occasionally the final 'A1' was missed by skipping straight from –k = ln(13/15) to k = 0.143: as this is a given answer, some additional working was required. Occasionally, the result was verified by substituting k = 0.143 and evaluating $15000e^{-0.143}$. This was treated as a special case and got 1 mark only. It is important candidates know the difference between 'show' and 'verify'.	e.g. $k = -\ln(13000/15000) = 0.143$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$15000e^{-0.143t} = 20000e^{-0.2t}$	M1*	must be correct, but could use a more accurate value for k	If M0, SCB1 for 5 – 5.1 years from correct calculations for each car, rot e.g. $t = 5$, £7358 (Brian), £7338 (Kate) or (£7334 with more accurate k)
		iii	$\Rightarrow (15000/20000) = e^{(0.143 - 0.2)t}$	M1dep	dep*	o.e. e.g. $\ln 15000 - 0.143t = \ln 20000 - 0.2t$
		iii	$\Rightarrow t = \ln 0.75 / -0.057 =$ 5.05 years so after 5 years	A1	cao accept answers in the range 5 – 5.1 **Examiner's Comments** This was a little more demanding, requiring candidates to combine the $e^{-0.2}$ and $e^{-0.143}$ terms. This defeated quite a few candidates. Some listed the values of each car for t = 1, 2, 3, etc years, and if successful picked out t = 5 as when they were closest. This was judged to be worth 1 mark only. Exponential growth and decay is well understood by most candidates, and this was a high scoring question.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	At P, $(e^x - 2)^2 - 1 = 0$			
		i	$\Rightarrow e^x - 2 = [\pm]1,$	M1	square rooting – condone no ±	
		i	$e^x = [1 \text{ or}]\ 3$			
		i	*or* $(e^x)^2 - 4e^x + 3 = 0$	M1	expanding to correct quadratic and solve by factorising or using quadratic formula	condone e^*x*^2
		i	$\Rightarrow (e^x - 1)(e^x - 3) = 0$, $e^x = 1$ or 3			
		i	$\Rightarrow x = [0 \text{ or}]\ \ln 3$	A1	*x*-coordinate of P is ln 3; must be exact **Examiner's Comments** Most candidates succeeded in finding x = ln 3, either by square rooting or solving thequadratic in e^x. The second method was somewhat compromised by setting x = e^x (ratherthan a different variable) to get a quadratic in x, though we condoned this for both marks.	condone P = ln 3, but not *y* = ln 3
		ii	$f'(x) = 2(e^x - 2)e^x$	M1	chain rule	e.g. 2 *u* × their deriv of e^x
		ii		A1	correct derivative	$2(e^x - 2)x$ is M0
		ii	= 0 when $e^x = 2$, $x = \ln 2$ *	A1	not from wrong working **NB AG**	or verified by substitution
		ii	*or* $f(x) = e^{2x} - 4e^x + 3$	M1	expanding to 3 term quadratic with $(e^x)^2$ or e^{2x}	condone e^*x*^2
		ii	$\Rightarrow f'(x) = 2e^{2x} - 4e^x$	A1	correct derivative, not from wrong working	
		ii	= 0 when $2e^{2x} = 4e^x$, $e^x = 2$, $x = \ln 2$ *	A1	or $2e^x(e^x - 2) = 0 \Rightarrow e^x = 2$, $x = \ln 2$	or verified by substitution
		ii			not from wrong working **NB AG**	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$y = f(\ln(2)) = -1$	B1	**Examiner's Comments** This provided a simple four marks for most candidates, using a chain rule to find the derivative, setting this to zero and solving to get x = ln 2. A neat alternative method was to recognise that the $(e^x - 2)^2$ term must be non-negative and minimum when $e^x - 2 = 0$, or x = ln 2.	
		iii	$\int_0^{\ln 3}[(e^x-2)^2-1]dx$ $=\int_0^{\ln 3}[(e^x)^2-4e^x+4-1]dx$	M1	expanding brackets must have 3 terms: $(e^x)^2 - 4$ is M0, condone e^*x*^2	or if $u = e^x$, $\int_1^3[u^2 - 4u + 4 - 1]/u\,du$
		iii	$=\int_0^{\ln 3}[e^{2x}-4e^x+3]dx$	A1	$\int e^{2x} - 4e^x + 3$ [dx] (condone no dx)	$= \int u - 4 + 3/u\,du$
		iii	$=\left[\frac{1}{2}e^{2x}-4e^x+3x\right]_0^{\ln 3}$	B1	$\int e^{2x} = ½\, e^{2x}$	$= [\,½\, u^2 - 4u + 3\ln u$
		iii		A1ft	$[½\, e^{2x} - 4e^x + 3x]$	
		iii	= (4.5 – 12 + 3ln3) – (0.5 – 4)			
		iii	= 3ln3 – 4 [so area = 4 – 3ln3]	A1	condone 3ln3 – 4 as final ans; mark final ans **Examiner's Comments** This proved to be a rather costly part for candidates unless they recognised the requirement to multiply out $(e^x - 2)^2 - 1$ to get $e^{2x} - 4e^x + 3$ and then integrate term-by-term. Other attempts using substitution or parts usually got nowhere. Although originally we required candidates to give the area as 4 – 3ln3, very few actually did this, so it was decided to condone a (negative) area of 3ln3 – 4.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$y = (e^x - 2)^2 - 1$ $x \leftrightarrow y$			
		iv	$x = (e^y - 2)^2 - 1$			
		iv	$\Rightarrow x + 1 = (e^y - 2)^2$	M1	attempt to solve for y (might be indicated by expanding and then taking lns)	or x if x and y not interchanged yet or adding (or subtracting) 1
		iv	$\Rightarrow \pm\sqrt{(x + 1)} = ey - 2$ (+ for $y \geq \ln 2$)	A1	condone no ±	
		iv	$\Rightarrow 2 + \sqrt{(x + 1)} = e^y$			
		iv	$\Rightarrow y = \ln(2 + \sqrt{(x + 1)}) = f^{-1}(x)$	A1	must have interchanged x and y in final ans	
		iv	Domain is $x \geq -1$	B1	must be $\geq$ and x (not y)	if not specified, assume first ans is domain and second range
		iv	Range is $y \geq \ln 2$	B1	or $f^{-1}(x) \geq \ln 2$, must be $\geq$ (not x or $f(x)$) if $x > -1$ and $y > \ln 2$ SCB1	
		iv		M1	recognisable attempt to reflect curve, or any part of curve, in $y = x$	$y = x$ shown indicative but not essential

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv		A1	good shape, cross on $y = x$ (if shown), correct domain and range indicated. [see extra sheet for examples] **Examiner's Comments** Rather more than half of the candidates managed the inverse function well, though a few made errors at the last stage of taking the square root, and concluded with y = ln($\sqrt{(x+1)}$)+ 2, or y = ln($\sqrt{(x+1)}$+ ln 2. Some were perhaps encouraged by the previous part to multiply out $(e^x - 2)^2$ again, though they could still obtain a method mark for a step towards finding y in terms of x. It was not uncommon to see candidates taking logs of individual terms.	e.g. – 1 and ln 2 marked on axes
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	$1 + k$	B1		**Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. This was an easy write-down for virtually all candidates, except those few who did not know that $e^0 = 1$.
		ii	$f'(x) = 2\,e^{2x} - 2\,k\,e^{-2x}$	B1		
		ii	$f'(x) = 0 \Rightarrow 2\,e^{2x} - 2\,k\,e^{-2x} = 0$	M1	their derivative = 0	
		ii	$\Rightarrow e^{2x} = k\,e^{-2x}$			
		ii	$\Rightarrow e^{4x} = k$, $4x = \ln k$, $x = ¼ \ln k$*	A1	NB AG	
		ii	$y = e^{(½ \ln k)} + k\,e^{(-½ \ln k)}$	M1	substituting $x = ¼ \ln k$ into $f(x)$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$= \sqrt{k} + k/\sqrt{k} = 2\sqrt{k}$	A1cao	or $2k^{1/2}$	**Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The first two marks were pretty universally earned, but deriving $x = ¼ \ln k$, together with the final 'A' mark for getting $2\sqrt{k}$, caused a few problems, with some inaccurate logarithm work. For example, $e^{1/2 \ln k} = 1/2\, k$ was a commonly seen misconception.
		iii	$\text{Area} = \int_0^{\frac{1}{2}\ln k} (e^{2x} + ke^{-2x})[\mathrm{d}x]$	B1	correct integral and limits (soi)	
		iii	$= \int_0^{\frac{1}{2}\ln k} (e^{2x} + ke^{-2x})\,\mathrm{d}x = \left[\frac{1}{2}e^{2x} - \frac{1}{2}ke^{-2x}\right]_0^{\frac{1}{2}\ln k}$	B1	$\left[\frac{1}{2}e^{2x} - \frac{1}{2}ke^{-2x}\right]$	
		iii	$= ½\, k – ½ – ½ + ½\, k$	M1	$e^{\ln k} = k$ or $e^{-\ln k} = 1/k$ (soi)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= k - 1$	A1		**Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The integration was usually correct, but, thereafter, as in part (ii), the simplification to arrive at $k - 1$ proved to be tricky, with similar errors being made.
		iv	(A) $g(x) = e^{2(x + \frac{1}{4}\ln k)} + ke^{-2(x + \frac{1}{4}\ln k)}$	M1	Substitute $x + \frac{1}{4}\ln k$ for x in $f(x)$	condone missing brackets
		iv	$= e^{2x}.e^{\frac{1}{2}\ln k} + k\,e^{-2x}.\,e^{-\frac{1}{2}\ln k}$	M1	$e^{p+q} = e^p \times e^q$ used	
		iv	$= (e^{\ln k})^{1/2}\,e^{2x} + k.(e^{\ln k})^{-\frac{1}{2}}\,e^{-2x}$			
		iv	$= k^{1/2}\,e^{2x} + k.k^{-1/2}.\,e^{-2x}$			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$= \sqrt{K}\,(e^{2x} + e^{-2x})$*	A1	**NB AG** – must show enough working	e.g. $k^{1/2}\ e^{2x} + k.k^{-1/2}.\ e^{-2x}$ **Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. Most attempts correctly substituted x + ¼ ln k for x in f(x) to gain the first M mark, but we needed to see clear evidence of how this simplifies to the given result. Often candidates seemed to be working backwards from this without really understanding the process.
		iv	(B) $g(-x) = \sqrt{k}\,(e^{-2x} + e^{2x})$	M1	substituting $-x$ for x	condone 'f' used instead of 'g' for M1

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	= g(x) so g is even	A1	must include g(–x) = g(x), and either define an even function or conclude that g is even	not f(–x) = f(x) for A1 **Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The definition of an even function was well known, but sometimes the structuring of the proof was indecisively presented. Some used 'f' instead of 'g' (here, f is indeed not an even function!), and we required to see either a clear statement of the definition of an even function, or a clear conclusion that g is therefore even. The structure 'g(–x) = … = … = g(…) ⇒ g is even' is the most transparent formulation to use in such proofs, rather than starting them by stating that g(–x) = g(x), viz the result they are trying to prove!
		iv	(C) g(x) is symmetrical about the y-axis, and	B1		
		iv	f(x) is g(x) translated ¼ ln k in x-direction	B1	allow 'shift' or 'move'	or g is f translated – ¼ ln k

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	so f(x) is symmetrical about $x = \frac{1}{4} \ln k$	B1	allow final B1 even if unsupported	or incorrectly supported **Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The argument here proved beyond most candidates, with only 20% getting full marks. Many stated that f was an even function, perhaps thinking that any line of symmetry sufficed. Sometimes it was indeed a little difficult to decide whether candidates were referring to f or g in their answers.
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
17			$fg(x) = \ln(2 + e^x)$	M1	condone missing brackets	
			$\Rightarrow \ln(2 + e^x) = 2x$			
			$\Rightarrow 2 + e^x = e^{2x}$	A1		
			$\Rightarrow e^{2x} - e^x - 2$ [= 0]	M1	Rearranging into a quadratic in e^x	may be implied from both correct roots
			$\Rightarrow (e^x - 2)(e^x + 1) = 0$, $e^x = 2, -1$	A1	obtaining roots 2, –1	–1 root may be inferred from factorising
			$\Rightarrow e^x = 2$, $x = \ln 2$	A1	$x = \ln 2$ only, not from ww	$x = \ln(-1)$ is A0 **Examiner's Comments** Virtually all candidates formed the composite function in the correct order to obtain $fg(x) = \ln(2 + e^x)$. A few then simplified this to $\ln 2 + x$ and therefore made no further progress. Of those who did correctly proceed to $2 + e^x = e^{2x}$, a substantial minority then incorrectly took logs of each side to reach $\ln 2 + x = 2x$. Of those who correctly rearranged the equation into a quadratic in e^x, nearly all then gained full marks, correctly rejecting the $e^x = -1$ solution.
			Total	5		

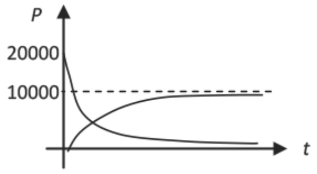
Question			Answer/Indicative content	Marks	Part marks and guidance	
18		a	$\log_{10} N = \log_{10} A + kt \log_{10} 2$	M1(AO1.1)		
			Equation above is of the form $y = mx + c$ [with $\log_{10} N$ as y and t as x]	E1(AO1.2)		
			Gradient = $k \log_{10} 2$	A1(AO2.2a)		
			Intercept = $\log_{10} A$	A1(AO2.2a)		
				[4]		
		b	$k \log_{10} 2 = 0.2 \Rightarrow k = 0.66[438]$	B1(AO1.1)		
			$\log_{10} A = 2 \Rightarrow A = 100$	B1(AO1.1)		
				[2]		
		c	$N = 100\times2^{0.66\ldots\times24} = 6\ 300\ 000$ FT their A, k	B1(AO3.4) [1]	Answer in range 5 860 000 to 6 400 000	
		d	E.g. the piece of bread may not be sufficient to support the number of bacteria	E1(AO3.5b) [1]	OR bacterial growth may obey different rules for large values of t	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance		
19			let $u=1+\sqrt{x} \quad du=\frac{1}{2}x^{-\frac{1}{2}}\,dx$ $\Rightarrow$ dx = 2 (u – 1)du $\Rightarrow \int_0^1 \frac{1}{1+\sqrt{x}}dx=\int_1^2 \frac{2(u-1)}{u}du$ $=\int_1^2\left(2-\frac{2}{u}\right)du$ $=\left[2u-2\ln u\right]_1^2$ = 4 – 2ln 2 – 2 = 2 – 2ln 2 or 2 – ln 4	M1(AO3.1a) A1(AO1.1) A1(AO1.1) M1(AO3.1a) A1(AO1.1) A1cao(AO1.1) [6]	substituting $u=1+\sqrt{x}$ or $w=\sqrt{x}$ dx = 2(u – 1) du or dx = 2w dw $\frac{2(u-1)}{u}(du)$ or $\frac{2w}{(w+1)}(dw)$ splitting fraction or dividing to get $2-\frac{2}{(w+1)}$ (or substituting u = w + 1 ⇒ $\frac{2(u-1)}{u}$ and then splitting fraction) or $[2w-2\ln(w+1)]_0^1$ if still in terms of w	Evidence of method must be seen Evidence of method must be seen	
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance	
20		a	$\log_3 x^2 a$	B1(AO1.1) [1]		
		b	$x^2 a = 3^2$ $x = [\pm]\frac{3}{\sqrt{a}}$ oe Disregard $x = -\frac{3}{\sqrt{a}}$ as x cannot be negative	M1(AO1.1) A1(AO1.1) A1(AO2.1) [3]	Must be clear that the negative root has been considered and disregarded	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
21		a	$-0.03e^{-0.03t}$	B1(AO1.2) [1]		
		b	Decreasing function because $e^{-0.03t}$ is positive [for all values of t] so the gradient is negative.	E1(AO2.2a) [1]	Explanation may include a sketch graph of the function $70e^{-0.03t}$ but it must be clear that the graph is of the function and the answer must clearly refer to the gradient of the function and not the trend in the data	
		c	A 70	B1(AO1.1) [1]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	B 38.[4168…]	B1(AO1.1) [1]		
		d	Data values decreasing so decreasing function is suitable At $t = 0$, calculated D = 70 and this matches the data At $t = 20$, data value is 40 which is not exact but close	E1(AO3.5a) B1(AO3.5a) B1(AO3.5b) [3]		
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
22		a	*A* 	M1(AO1.1) M1(AO1.1)	P_G shape through O P_R shape through (0, 20000), [condone graphs for –ve *t*]	
		a	*B* asymptote for P_G = 10 000 Asymptote for P_R = 0	A1(AO2.2a) A1(AO2.2a) [4]	Or p = 10 000 Or p = 0	
		b	Red squirrels zero Grey 10 000	B1(AO3.4) B1(AO3.4) [2]		
		c	One relevant comment evaluating the validity of the model	B1(AO3.5a)	E.g. One of • Grey population increases as would be expected [since grey squirrels are larger and more successful] • Red population decreases as would be expected [since red	

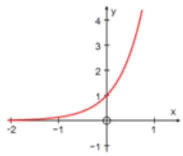
Question			Answer/Indicative content	Marks	Part marks and guidance		
				[1]	squirrels have to compete with the larger grey squirrels for food] • Number of squirrels tends to a limit as would be expected [since there is limited food and space] • Would expect grey population to grow slower at first • Would expect red population to fall slower at first		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	$\frac{dP_G}{dt} = 10\ 000ke^{-kt}$ $\frac{dP_R}{dt} = -20\ 000ke^{-kt}$ so $\frac{dP_R}{dt} = -2\frac{dP_G}{dt}$	M1(AO3.1b) A1(AO1.1) A1(AO1.1) E1(AO2.1) [4]	Attempts to differentiate either or both Or in words	
		e	$10\ 000\ (1 - e^{-3k}) = 20\ 000\ e^{-3k}$ $\Rightarrow 1 - e^{-3k} = 2\ e^{-3k}$ $\Rightarrow e^{-3k} = \frac{1}{3}$ $\Rightarrow -3k = \ln\left(\frac{1}{3}\right)$ $\Rightarrow k = -\frac{1}{3}\ln\left(\frac{1}{3}\right) = 0.366$ or $\frac{1}{3}\ln 3$	M1(AO1.1a) A1(AO1.1) M1(AO1.1) A1cao(AO2.1) [4]	Taking natural logs of both sides	
			Total	15		

Question			Answer/Indicative content	Marks	Part marks and guidance	
23			$\sum_{r=1}^{4} \ln \frac{r}{r+1} = \ln\frac{1}{2} + \ln\frac{2}{3} + \ln\frac{3}{4} + \ln\frac{4}{5}$ = ln 1 – ln 2 + ln 2 – ln 3 + ln 3 – ln 4 + ln 4 – ln 5 = 0 – ~~ln 2~~ + ~~ln 2~~ – ~~ln 3~~ + ~~ln 3~~ – ~~ln 4~~ + ~~ln 4~~ – ln 5 = –ln 5	B1(AO1.1) M1(AO3.1a) E1(AO2.2a) [3]	soi use of $\ln\left(\frac{a}{b}\right)$= ln a – ln b or ln a + ln b = ln ab $\ln\left(\frac{1}{2}\times\frac{2}{3}\times\frac{3}{4}\times\frac{4}{5}\right)=\ln\frac{1}{5}$ = ln 1 – ln 5= – ln 5 AG	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
24		a	(i) $f'(x) = e^{-x}\cos x - e^{-x}\sin x$ $f'(x) = 0$ and $e^{-x} \neq 0 \Rightarrow \cos x = \sin x$ $\Rightarrow \tan x = 1$ $\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$	M1(AO3.1a) A1(AO1.1) E1(AO2.2a) M1(AO1.1) A1(AO1.1)	product rule correct Use of $\frac{\sin}{\cos} = \tan$ $x = \frac{\pi}{4}$ (condone 45°) $\frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \ldots$	
		a	So an AP with $d = \pi$ (ii) $y = \frac{\sqrt{2}}{2}e^{-\frac{\pi}{4}}, -\frac{\sqrt{2}}{2}e^{-\frac{5\pi}{4}}, \frac{\sqrt{2}}{2}e^{-\frac{9\pi}{4}}, -\frac{\sqrt{2}}{2}e^{-\frac{13\pi}{4}}$ This is a GP with $r = -e^{-\pi}$	E1FT(AO2.1) M1(AO3.10a) A1(AO1.1) E1FT(AO2.1) [9]	must state the common difference substituting one value of x into $f(x)$ must state common ratio, www	FT their values of x FT their values of y
		b	Yes with explanation that values of x would continue to be separated by pi and so values of y would continue to have same common ratio.	E1(AO2.2a) [1]		
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance		
25		a	$c = 4.45$	B1(AO3.3) [1]			
		b	$\log_{10}y = -0.37 \log_{10}t + 4.45$ $y = 10^{-0.37 \log_{10} t + 4.45}$ $\log_{10}t^{-0.37}$ seen $10^{4.45} \times t^{-0.37}$ 21183.829.. ≈ 28 200 so $y \approx 28\ 200\ t^{-0.37}$	M1(AO2.1) M1(AO1.1) A1(AO1.1) [3]	may be awarded after combining logarithms AG		
		c	18 781 is close to 18776	B1(AO3.5a) [1]	BC	NB $t = 3$	
		d	**A** 12 507 or 12 508	B1(AO3.4) [1]	BC to 3, 4 or 5 s.f.		
		d	**B** 8986	B1(AO3.4) [1]	BC		
		e	Answer to A interpolation so more likely to be reliable Answer to B extrapolation beyond 2015 so unreliable	B1(AO3.5a) B1(AO3.5b) [2]			
			Total	9			

Question			Answer/Indicative content	Marks	Part marks and guidance	
26		a		G1(AO 1.2) G1(AO 1.2) [2]	Correct shape with x -axis as asymptote (0, 1) clearly shown	
		b	Stretch in the x-direction Stretch scale factor $\frac{1}{2}$	B1(AO 1.1a) B1(AO 1.1a) [2]	'Stretch' must be seen at least once for any marks to be awarded, but the word needn't be repeated	
		c	$\frac{dy}{dx} = 2e^{2x}$ When $x = 3$, $\frac{dy}{dx} = 2e^6$ and $y = e^6$ Tangent is $y - e^6 = 2e^6\,(x - 3)$ $y = 2e^6x - 5e^6$ $y = e^6\,(2x - 5)$	M1(AO 1.1a) A1(AO 1.1b) A1(AO 1.1a) B1(AO 1.1a) M1(AO 1.1a) A1(AO 2.1) [6]	Allow for any ke^{2x} Correct $2e^{2x}$ Allow method if 403 or better used for e^6 Must be in this form	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
27			Sequence of the form ln$a$, lnar, lnar^2 …	M1(AO 2.5)	Use of notation for geometric sequence	
			ln$a$, ln$a$ + ln$r$, ln$a$ + 2ln$r$…	M1(AO 3.1a)	Laws of indices used	
			Arithmetic sequence with common difference ln r	E1(AO 2.2a)	Laws of indices used	
				[3]		
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
28		a	$k = 5$	B1(AO 3.3) [1]	$36 = 9k - 9$	
		b	$x = 7.5$, model gives 140.25 ≈ 140 so model works well	B1(AO 3.4) B1(AO 3.5a) [2]		
		c	*A* $V = 88.5$ substitution of $x = 5.20$ and 5.21 to obtain 88.33 and 88.58 respectively	B1(AO 3.1a) B1(AO 1.1) [2]	NB 88.496 878236	
		c	*B* $\frac{dV}{dt} = 28e^{-0.2t}$ $= 10.3$ when $t = 5$ $\frac{dV}{dt} = (10x - x^2) \times \frac{dx}{dt}$ $\frac{dx}{dt} = \frac{10.3}{10 \times 5.21 - 5.21^2} = 0.413$ [cms^{-1}]	M1(AO 1.1) A1(AO 2.1) M1(AO 3.4) A1(AO 1.1) [4]	NB 10.300 6243528 Accept awrt 0.412 or 0.413	
		c	*C* Forever for a full glass or more than 70 cm^3 full after 5 seconds from (*A*) so better to fill the two half glasses	B1(AO 3.5b) B1(AO 3.4) [2]		
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance		
29		i	curve of increasing gradient in 1st and 2nd quadrant which does not cut x-axis but tends towards it in 2nd quadrant	M1	M0 if curves up in 2nd quadrant or back in 1st quadrant	condone touching x-axis	
			through (0, 1)	A1	intercept may be identified in supporting commentary or on graph	condone axes not labelled	
				[2]			
					Examiner's Comments Most candidates scored full marks with this part of the question, although the quality of the sketches were variable. A few drew $y = 2x$ or $y = x^2$, and some candidates marked the y -intercept as (0, 2), losing an easy mark.		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	$\log_a\left(\frac{x^5 \times 6}{2x}\right)$ oe	B1	**NB** $\log_a (3x^4)$ may be embedded in combining of all terms on RHS	condone omission of base	
			correct attempt to remove logs on both sides	M1	**NB** $\log_a (3a^3x^4)$ eg $w = a^{3 + \log_a x^5 - \log_a 2x + \log_a 6}$ may follow incorrect combination of log terms	condone omission of base, may be awarded before **B1**	
			$[w =]3a^3x^4$ cao	A1 **[3]**	**Examiner's Comments** Over half the candidates failed to score on this question, with difficulties seen by candidates attempting to combine the logarithms successfully. In attempting to make w the subject, candidates sometimes "divided by $\log_a$" or raised both sides to the power 10, and only a minority earning the method mark.		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance	
30		i	Initial temperature is 10.5 [°C] boiling point is 80[°C]	B1 B1 [2]	**Examiner's Comments** The initial temperature was almost always correct, but the boiling point was sometimes incorrect or missing, suggesting that the limit of e^{-kt} as t tends to infinity was not known.	
		ii	$30 = 10.5 + 69.5(1 - e^{-k})$, $\Rightarrow e^{-k} = 1 - 19.5/69.5$ $\Rightarrow -k = \ln(0.7194\ldots)$ $\Rightarrow k = -\ln(0.7194\ldots) = 0.3293\ldots$	B1 M1 A1 [3]	re-arranging and taking lns (correctly) art 0.33 or ln(139/100) o.e. **Examiner's Comments** Exponential growth and decay equations are usually well answered, and this was no exception, with most candidates scoring full marks.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		iii	$79 = 10.5 + 69.5(1 - e^{-kt})$ $\Rightarrow e^{-kt} = 1 - 0.9856.. = 0.014388489\ldots$ $\Rightarrow t = -\ln(0.014388\ldots)/0.3293$ [= 12.879…]☐ = 13 mins	M1 M1 A1cao [3]	substituting θ = their (80 –1) into the eqn and rearranging for e^{-kt} taking lns correctly	Trial and error: e.g. t = 12, θ = 78.66 t = 13, θ = 79.04, so 13 mins SCB2	
					Examiner's Comments This question depended on the boiling point being correct, so the facility was lower. However, over half the candidates got full marks. The solution was made considerably harder if an inequality was used, as the working needed to show the reversing of the inequality signs.		
			Total	8			

Question			Answer/Indicative content	Marks	Part marks and guidance	
31		a	$A = 22 \text{ ms}^{-1}$	B1(AO1.1b)	from $t = 0$	
			$25.3 = 22 + B(1 - e^{-0.2\times2})$	M1(AO3.3)		
			$B = 10$	A1(AO1.1b)	NB from 10.0097…	
				[3]		
		b	$22 + 10(1 - e^{-0.2\times5})$	M1(AO3.4)		
			28.3(212…) so good fit	A1(AO3.5a)		
				[2]		
		c	$-10 \times (-0.2) \times e^{-0.2t}$ oe soi	M1*(AO3.1a)		
			substitution of $t = 5$ in *their* derivative	M1dep(AO1.1b)		
			0.74 ms^{-2}	A1(AO1.1b)	0.73575888…to 2 sf or better	
				[3]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	as $t \to \infty$ oe	M1(AO3.5a)		
			$V \to 32$	A1(AO3.4)		
			$32 \times 3.6 = 115.2$ kmh^{-1}	B1(AO1.1b)	FT their long term V	
			$115.2 < 121$ so not 10% above the speed limit	E1(AO3.2a)	or 5.2 is 4.7% of 110	
				[4]		
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
32			$\dfrac{\ln x \times 2 - 2x \times \frac{1}{x}}{(\ln x)^2}$	M1(AO3.1a)	Quotient Rule allow sign errors only	or Product & Chain Rules
				A1(AO1.1b)	numerator correct	$2 \times \dfrac{1}{\ln x} +$ $2x \times (-1) \times (\ln x)^{-2} \times \frac{1}{x}$
				A1(AO1.1b)	denominator correct	
			indicates $\frac{dy}{dx} = 0$	M1(AO2.1)		
			x = e	A1(AO2.4)		
			and y = 2e shown www	A1(AO3.1a)	AG	
			either finds second derivative *or* considers *their* gradient function either side of x = e	M1(AO2.1)		
			2nd derivative is $\dfrac{4 - 2\ln x}{x(\ln x)^3}$	A1(AO1.1)		
			$\frac{2}{\mathrm{e}}$ $(0.7357588..) > 0$ so minimum or signs of gradient function either side of x = e identified and explained	E1(AO2.4) [9]	Allow FT if previous M1 awarded, but must be correctly evaluated	Allow decimal equivalent to 2 sf or better
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
33		a	$\frac{dx}{dt}$ is the rate at which x is increasing Mass of B is x, so mass of A is $(1 - x)$ $\frac{dx}{dt} \propto x(1-x)$, so $\frac{dx}{dt} = kx(1-x)$	B1(AO2.1) [1]	Must indicate where terms come from AG	
		b	$\int \frac{1}{x(1-x)}\,dx = \int k\,dt$ $\frac{1}{x(1-x)} = \frac{A}{x} + \frac{B}{1-x}$ 1 &≡ A(1 – x) + Bx &⇒ A = 1, B = 1 $\int\left(\frac{1}{x}+\frac{1}{1-x}\right)dx = \int k\,dt \Rightarrow \ln x - \ln(1-x) = kt + c$ $t = 0,\ \ x = 0.2 \Rightarrow c = \ln\frac{1}{4}$ (oe) $\frac{4x}{1-x} = e^{kt}$ (oe) $x = \frac{e^{kt}}{4+e^{kt}}$	M1(AO3.1a) M1(AO3.1a) A1(AO2.2a) M1*(AO1.1a) M1dep*(AOs3.1a) M1dep*(AO1.1b) A1(AOs2.5) [7]	Separation of variables Find partial fractions (may be implied) Condone sign error, but must have two ln terms and +c Use of initial conditions; may be done after equation is rearranged Rearrange equation to remove logs; may be done before finding c oe, but must be of the form x = f(t)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	$t = 15$, $x = 0.9$ &⇒ $36 = e^{15k}$ (oe) $k = 0.239$ to 3sf	M1(AO3.1) A1(AO1.1b) [2]	Substitute values in their solution	
		d	$t = 30 \Rightarrow$ mass of B is $\frac{e^{0.239\times30}}{4+e^{0.239\times30}} = 0.997$ kg	B1(AO3.4) [1]		
		e	As $t \to$ ro, $x \to 1$ and so $1 - x \to 0$, so the model predicts there is a very small amount of A remaining when t is large	B1(AO3.5a) [1]	May evaluate x for large t (eg $t = 100$)	
			Total	12		
34			$\sqrt{1+\ln x} = 0 \Rightarrow \ln x = -1$ $x = \frac{1}{e}$	M1(AO1.1a) A1(AO2.2a) [2]	oe but must be exact value	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
35		a	4	B1(AO1.2) [1]	**Examiner's Comments** Many candidates gave the correct answer to this part. However, some candidates omitted this, and some others gave answers in terms of *a*.	
		b	–1	B1(AO1.1) [1]	**Examiner's Comments** This part was sometimes omitted, and some candidates gave answers involving *a*.	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
36		a	$A = 500$ $k = 1.044$	B1(AO3.3) B1(AO1.1) [2]	**Examiner's Comments** While many candidates wrote down the correct answers from the information given, quite a large minority wrongly calculated the value of k that would make the population 650 after 10 days.	
		b	500×1.044^{10} = 769.1 which is not close to 650 so not consistent	M1(AO3.4) A1(AO1.1) [2]	Ft www **Examiner's Comments** Candidates who had used the value of 650 to find their value of k could gain only partial credit for this part; the common wrong value of k suggests that the model is consistent. Many candidates correctly found that Munirah's first model would give about 769 flies after 10 days, which is not consistent with 650.	May use 650 and show k or A different

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	exponential growth oe	B1(AO3.5a) [1]	OR increase for ever, oe **Examiner's Comments** Almost all candidates realised that the number of flies would continue to rise in the long run.	
		d	$t = 0$, $N = 500$ $t = 10$, $N = 650.37 \approx 650$ model predicts number of fruit flies tends to 750 in the long run	B1(AO1.1) B1(AO1.1) B1(AO3.5a) [3]	Oe 'N will not go beyond 750' **Examiner's Comments** Although there were many completely correct answers, a lot of candidates identified only one or two of the ways. Some of these candidates used the increase of 4.4% per day from Munirah's original model to suggest an answer; this is not valid. Other candidates made far more general comments that did not answer the question.	Allow shown using large value of t

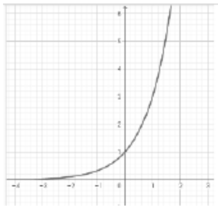
Question			Answer/Indicative content	Marks	Part marks and guidance	
		e	(i) Oscillations	B1(AO3.5b) [1]	oe **Examiner's Comments** Very few candidates identified the required feature as (decreasing) oscillations. A considerable number of candidates did not attempt this question.	
			(ii) cosine or sine oe	B1(AO3.5c) [1]	Eg Introducing a multiplier of $(-1)^t$ if t is a large integer. **Examiner's Comments** The most common response was to omit this part. The incorrect answer of a logarithmic function was considerably more common than a correct answer; correct answers invariably involved mention of a sine or cosine function.	
			Total	10		

Question	Answer/Indicative content	Marks	Part marks and guidance
37	Curve crosses the x-axis when $y = 0$ $y = (k - x)\ln x = 0$	M1 (AO 3.1a)	Attempt to solve $y = 0$
	Either $k - x = 0$ or $\ln x = 0$ $x = k$ or 1	A1 (AO 1.1b)	Both roots required
	EITHER $\text{Area} = \int_1^k (k - x)\ln x \, \mathrm{d}x$ Let $u = \ln x,\ \frac{\mathrm{d}v}{\mathrm{d}x} = k - x,\ \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x},\ v = kx - \frac{1}{2}x^2$	M1 (AO 2.1)	Using integration by parts with $u = \ln \frac{\mathrm{d}v}{\mathrm{d}x} = k - x$ x, clearly argued
	$\text{Area} = \left[\left(kx - \frac{1}{2}x^2\right)\ln x\right]_1^k - \int_1^k \frac{1}{x}\left(kx - \frac{1}{2}x^2\right) \mathrm{d}x$	A1 (AO 1.1b)	
	$\left[\left(kx - \frac{1}{2}x^2\right)\ln x\right]_1^k - \int_1^k \left(k - \frac{1}{2}x\right) \mathrm{d}x$	M1 (AO 3.1a)	
	$\left[\left(kx - \frac{1}{2}x^2\right)\ln x - \left(kx - \frac{1}{4}x^2\right)\right]_1^k$	A1 (AO 1.1b)	Allow without limits
	$\left(\left(k^2 - \frac{1}{2}k^2\right)\ln k - \left(k^2 - \frac{1}{4}k^2\right)\right)$ $-\left(\left(k - \frac{1}{2}\right)\ln 1 - \left(k - \frac{1}{4}\right)\right)$	M1dep (AO 1.1a)	Simplifying the integrand
	$= \frac{1}{2}k^2 \ln k - \frac{3}{4}k^2 + k - \frac{1}{4}$	A1 (AO 1.1b)	Second part correct
	OR Integral split into two separate integrals $\int_1^k k \ln x \, \mathrm{d}x$	[8]	Using limits. Depe ndendent on M mark for integration by parts

Question			Answer/Indicative content	Marks	Part marks and guidance		
				M1	Cao		
			Let $u=\ln x,\quad \frac{dv}{dx}=k,\quad \frac{du}{dx}=\frac{1}{x},\quad v=kx$				
			$= \left[kx\ln x\right]_1^k - \int_1^k \frac{1}{x}kx\ dx$	M1	Using integration by parts with u = ln x, $\frac{dv}{dx}=k$ or u = ln x, $\frac{dv}{dx}=\pm x$ clearly argued		
			$\left[kx\ln x\right]_1^k - \int_1^k k\ dx = \left[kx\ln x - kx\right]_1^k$ (k^2 ln k – k^2) – (k ln 1 – k) = k^2 ln k – k^2 + k And $\text{Area} = \int_1^k x\ln x\ dx$	M1dep	Simplifying the integrand		
			Let $u=\ln x,\quad \frac{dv}{dx}=x,\quad \frac{du}{dx}=\frac{1}{x},\quad v=\frac{1}{2}x^2$ $= \left[\frac{1}{2}x^2\ln x\right]_1^k - \int_1^k \frac{1}{x}\times\frac{1}{2}x^2 dx$	A1			
			$\left[\frac{1}{2}x^2\ln x\right]_1^k - \int_1^k \frac{1}{2}x\ dx$ $\left[\frac{1}{2}x^2\ln x - \frac{1}{4}x^2\right]_1^k$ $\left(\frac{1}{2}k^2\ln k - \frac{1}{4}k^2\right) - \left(\frac{1}{2}\ln 1 - \frac{1}{4}\right) = \frac{1}{2}k^2\ln k - \frac{1}{4}k^2 + \frac{1}{4}$ $\text{Area} = \left(k^2\ln k - k^2 + k\right) - \left(\frac{1}{2}k^2\ln k - \frac{1}{4}k^2 + \frac{1}{4}\right)$	A1	Substituition of limits seen in at least one integral. Dependent on M mark for integration by parts		
			$= \frac{1}{2}k^2\ln k - \frac{3}{4}k^2 + k - \frac{1}{4}$	A1			

Question			Answer/Indicative content	Marks	Part marks and guidance		
					Both integrals correct at this stage Allow without limits Both integrals fully correct Allow without limits Cao		
					Examiner's Comments This is an exemplar of a question requiring an extended answer – there are 4 method marks in the scheme. Candidates had to structure their answer. Had a value for k been given, the definite integral would have been possible on many calculators and the question may have become a "detailed reasoning" question. Successful candidates generally used integration by parts with $\frac{dv}{dx} = k - x$. It needed much more work to expand the bracket and split the integral in two. Some candidates lost the final mark, as they did not tidy up their answer so had too many terms.		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					AfL In an unstructured question like this one, do not give up or leave it blank because you do not know how to calculate the limits. Find the indefinite integral to make sure of 4 out of 8 marks. Using incorrect limits could also have been credited a method mark.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
38		a	(i) 	B1 (AO 1.2) [1]	correct shape in both quadrants | condone touching the *x*-axis, but not cutting it **Examiner's Comments** Sketches should make clear that the function tends towards $y = 0$ but does not cut the x-axis	
			(ii) (0, 1)	B1 (AO 1.1) [1]	do not allow just $y = 1$ **Examiner's Comments** A clear indication of the coordinates (0,1) was required and not just $y = 1$.	
		b	(f(x) =)$\log_3 x$	B1 (AO 1.1) [1]	allow $\frac{\log x}{\log 3}$ eg **Examiner's Comments** There was an expectation that candidates would use base 3 logarithms, but a variety of correct functions were given and gained credit.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
39		a	$C = 2$ $A = 62$ $B = 10$	**B1 (AO 3.3)** **B1 (AO 3.3)** **B1 (AO 1.1)** **[3]**	since max when $t = 2$ since max when $(t - 2)^2 = 0$ from substitution of 22, 62 and 2 <u>Examiner's Comments</u> Candidates who did well in this question recognised that the maximum value of p has to be 62, and that this occurs when $t = 2$, thus obtaining A and C. The value of B soon follows. This who did less well wrote down three equations in three variables and often went astray.	
		b	substitution of 0.75 in $p = 62 - 10(t - 2)^2$ 46	**M1 (AO 3.4)** **A1 (AO 1.1)** **[2]**	FT *their* 2, 62, 10 allow 46.375 rounded to 2 or more sf <u>Examiner's Comments</u> Candidates who did well made the correct substitution in their formula. Candidates who did less well substituted $t = 45$.	

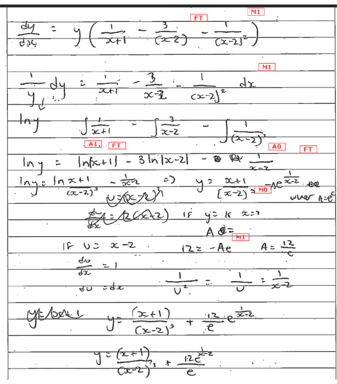
Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	*their* $62 - 10(t-2)^2 = 0$ [t =] 4 hours 29 minutes or 4 hours 30 minutes	M1 (AO 3.4) A1 (AO 2.4) [2]	or $\geq$ or $>$ 0 for **M1** NB $t = 2 + \sqrt{6.2}$ allow 4.49 or 4.5 [hours] <u>Examiner's Comments</u> Candidates who did well understood that the value of p couldn't be negative and obtained a value for t accordingly. Candidates who did less well worked with < instead of = or >.	
		d	substitution of t = 1, 3 and 5 awrt 59.4 ≈ 59 awrt 83.8 ≈ 84 awrt 88.8 ≈ 89	M1 (AO 3.4) A1 (AO 3.5a) [2]	or awrt 59.4, 83.8 and 88.8 found and supporting comment made eg they are ap proximately the same as the values in the table | if **M0** allow **SC1** for two values correctly found and shown to be consistent or supporting comment made <u>Examiner's Comments</u> Candidates who did well made correct substitutions and provided a supporting comment. Candidates who did less well neglected to substitute all three values or made no comment on what their calculations showed.	

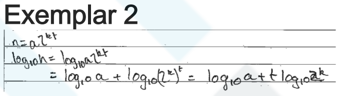
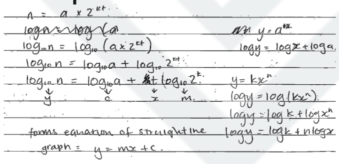
Question			Answer/Indicative content	Marks	Part marks and guidance		
		e	$p \to 90$ as $t \to$ large or when $t = 12$ $p = 89.99539\ldots$rounded to 2 or more sf comparison with value of p for $t = 5$ eg model predicts $p = 89$ for $t = 5$ and $p = 90$ for $t = 12$ so not good advice	B1 (AO 3.5a) B1 (AO 3.5a) [2]	or model predicts $p = 90$ for (any) $t \geq 7$ so not good advice **Examiner's Comments** Candidates who did well evaluated p at $t = 12$ and either at $t = 5$ or a value in between 5 and 12 and commented appropriately. Candidates who did less well made no supporting calculations but supplied a comment, or vice versa.	allow equivalent comment on 7 hours work for one extra mark	
			Total	11			

Question			Answer/Indicative content	Marks	Part marks and guidance	
40		a	$\frac{A}{(x+1)}+\frac{B}{(x-2)}+\frac{C}{(x-2)^2}$	B1 (AO 3.1a)	may be seen later	
			$x^2 - 8x + 9 = A(x-2)^2 + B(x + 1)(x-2) + C(x + 1)$	M1 (AO 2.1)		
			$A = 2$	A1 (AO 1.1)		
			$B = -1$	A1 (AO 1.1)	$\frac{2}{(x+1)}-\frac{1}{(x-2)}-\frac{1}{(x-2)^2}$	
			$C = -1$	A1 (AO 1.1)		
				[5]	**Examiner's Comments** Candidates who did well recognised the correct form of partial fractions and were able to work successfully to find the coefficients. Candidates who did less well made algebraic slips in clearing the fractions or made slips in arithmetic when finding the coefficients.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	$\int \frac{\mathrm{d}y}{y} = \int \frac{x^2-8x+9}{(x+1)(x-2)^2}\,\mathrm{d}x$ soi	M1* (AO 3.1a)	allow omission of integral signs and/or omission of d*y* and/or d*x*	
			use of their partial fractions in integration	M1* (AO 2.1)		
			$\ln\lvert y\rvert = 2\ln\lvert x+1\rvert - \ln\lvert x-2\rvert + \frac{1}{x-2} + c$	A1 (AO 1.1)	allow one sign error and/or one coefficient error	condone use of brackets instead of modulus signs; these two **A** marks are only available following the award of both **M** marks
				A1 (AO 1.1)	**A1** for any correct natural log integral on RHS FT *their* $\frac{2}{x+1}$ or *their* $\frac{-1}{x-2}$	
			substitution of $y = 16$ and $x = 3$	M1dep* (AO 1.1)	A1 $\frac{1}{x-2}$ FT for *their* $\frac{k}{(x-2)^2}$	
			correctly exponentiate both sides of their equation	M1 (AO 1.1)	expression must include + *c* and must include at least one natural log term; may be awarded after exponentiating	may be awarded following collection of like terms, which may contain errors NB *c* = – 1
			$y = \frac{(x+1)^2}{x-2}e^{\frac{3-x}{x-2}}$ oe	A1 (AO 2.1)		
				[7]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					eg $\frac{(x+1)^2}{x-2}e^{\frac{1}{x-2}}\ e^{-1}$ **Examiner's Comments** Candidates who did well recognised the need to use their result from part (a). They separated the variables successfully and were then able to integrate and substitute the values of *x* and *y* to find the constant of integration. Candidates who did very well were able to go on and find a correct expression for *y*. Candidates who did less well rearranged incorrectly when they attempted to separate the variables, or were unable to integrate the quadratic term correctly. They made slips in exponentiating both sides of their equation, usually assuming that the operation is distributive. Exemplar 4	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					 $\frac{dy}{dx} = y\left(\frac{1}{x+1} - \frac{3}{(x-2)} - \frac{1}{(x-2)^2}\right)$ FT M1 $\frac{1}{y}dy = \frac{1}{x+1} - \frac{3}{x-2} - \frac{1}{(x-2)^2}\,dx$ M1 $\ln y \quad \int\frac{1}{x+1} - \int\frac{3}{x-2} - \int\frac{1}{(x-2)^2}$ A1 FT A0 FT $\ln y = \ln\lvert x+1\rvert - 3\ln\lvert x-2\rvert - \frac{1}{x-2}$ $\ln y = \ln\frac{x+1}{(x-2)^3} - \frac{1}{x-2} \Rightarrow y = \frac{x+1}{(x-2)^3} - Ae^{\frac{1}{x-2}}$ M0 where $A = e^c$ if $y = 16$ $x = 3$ M1 if $u = x-2$ $\quad 12 = -Ae \quad A = \frac{12}{e}$ $\frac{du}{dx} = 1$ $du = dx \quad \frac{1}{u^2} = \frac{1}{u} = \frac{1}{x-2}$ $y = \frac{(x+1)}{(x-2)^3} + \frac{12}{e}e^{\frac{1}{x-2}}$ $y = \frac{(x+1)}{(x-2)^3} + \frac{12e^{\frac{1}{x-2}}}{e}$ In this response FT marks have been credited for the use of their partial fractions and separation of variables. One A mark has been credited FT, but the integration of the quadratic term went astray. The exponentiation of both sides was incorrect, but in spite of this, the method mark for substitution was subsequently earned.	
			Total	12		

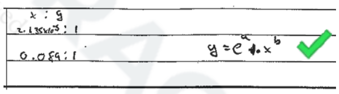
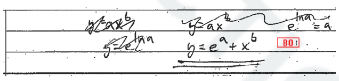
Question			Answer/Indicative content	Marks	Part marks and guidance	
41		a	$\log_{[10]}n = \log_{[10]}a + kt\log_{[10]}2$	M1(AO 1.1a)	AG Allow $t\log 2^k$	
			This is of form $y = mx + c$ [with $\log_{[10]}n$ as y and t as x]	E1(AO 1.1)		
				[2]	<u>**Examiner's Comments**</u> Most candidates could manage the use of logs and rearranging but quite a number did not finish off and explain why it was a straight line. **Exemplar 2**  $n = a2^{kt}$ $\log_{10}n = \log_{10}a2^{kt}$ $= \log_{10}a + \log_{10}(2^k)^t = \log_{10}a + t\log_{10}2^k$ The candidate correctly rearranges but does not relate their equation to $y = mx + c$ so only scores 1. **Exemplar 3**  $n = a \times 2^{kt}$ ~~logn = log (a~~ $\log_{10}n = \log_{10}(a \times 2^{kt})$ $\log_{10}n = \log_{10}a + \log_{10}2^{kt}$ $\log_{10}n = \log_{10}a + t\log_{10}2^k$ $y \quad c \quad x \quad m$ forms equation of straight line graph = $y = mx + c$. $y = kx^n$ $\log y = \log x + \log a$ $\log y = \log(kx^n)$ $\log y = \log k + \log x^n$ $\log y = \log k + n\log x$ The candidate makes doubly sure of the final mark by saying it is $y = mx + c$ and showing which parts of their equation correspond to m, c etc.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	Reasonable line of best fit drawn (by eye) Suitable method leading to a value eg use of intercept leading to $0.9 < \log a < 1.2$ So $7.4 < a < 15.85$ Suitable method leading to k value eg $k\log_{10}2$ = gradient ≈ 0.33 k in range $0 < k < 1.25$ and a in range $7.4 < a < 15.85$	B1(AO 1.1a) M1(AO 2.2a) M1(AO 1.1) A1(AO 2.2a) **[4]**	With $0.9 < c < 1.2$ Finding gradient of line or sub'n of t and log n	May use 2 points from line or condone use of 2 given points If gradient of exactly 1/3 used $k = 1.107\ 30936\ldots$
					<u>Examiner's Comments</u> Examiners were surprised how few actually drew in a line of best fit. There is the possibility that candidates drew their line of best fit on the question paper, rather than the answer booklet. Candidates should be reminded that it is only the answer booklet that is seen by the examiner, therefore any rough work, sketches, diagrams or annotations that are drawn on the question paper will not gain credit. Af L Choosing two of the points to determine the gradient is not generally the best	

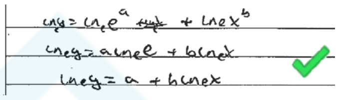
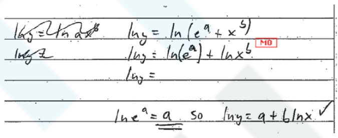
Question			Answer/Indicative content	Marks	Part marks and guidance	
					method but finding the gradient of their line is. It is possible that a candidate may have determined that all the points were generally close to the "line". This assumption was not penalised in this assessment.	
		c	$500000 = 10 \times 2^{1.1t}$ $1.1t \log 2 = \log 50000$ $t = 14.2$ $t = 14$ is 1/3/18 So 1/4/18	M1(AO 3.4) A1(AO 1.1) M1(AO 3.4) A1(AO 3.2a) [4]	Correct substitution Value of t (FT their a and k) — For $k = 1.10730936…$ $t = 14.1$ Same answer Translation into date Rounding up **<u>Examiner's Comments</u>** Many started well but many candidates did not correctly translate their t into the correct date or round up. Af L One of the features of the reformed qualification criteria is the increased emphasis on modelling. Candidates should ensure that their final answer reflects the context of the original problem and not only focus on the mathematical techniques.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	Suitable reason e.g. The data are only for a short time scale and cannot extrapolate e.g. There will not be enough people for the growth to continue	E1(AO 3.5b) [1]	**Examiner's Comments** A suitable reason was generally supplied for this part, even if the previous part was not correct. Af L Candidates should be reminded that subsequent questions related to the model may not require a successful answer to previous parts using the model. Good exam practice involves reading the full question before answering the individual parts, especially if the first parts appear challenging.	
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance		
42		a	$y = \dfrac{e^x}{1 - e^x}$ $y(1 - e^x) = e^x$ $[y = e^x(1+y)]$ $e^x = \dfrac{y}{1+y}$ $f^{-1}(x) = \ln\left(\dfrac{x}{1+x}\right)$	M1 (AO 1.1a) A1(AO 1.1) A1(AO 2.1) **[3]**	Clearing fractions Expression for e^x Condone 'y = ' Condone no brackets or mod **Examiner's Comments** Most candidates were well prepared for this standard question. Errors seen tended to be in forgetting to interchange x and y.	x and y may be inte rchanged at any stage	
		b	$f^{-1}(x) \neq 0$	B1 (AO 1.2) **[1]**	Allow $y \neq 0$ but not $x \neq 0$ **Examiner's Comments** The majority of candidates correctly identified that the range of $f^{-1}(x) \neq 0$		
			Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance	
43		a	$y = kx^n$	B1 (AO3.3) [1]	Allow any letters used for constant of proportionality and power. **Examiner's Comments** Many candidates seemed unprepared for this question which is covered by the specification point Ma14. Some tried to work backwards from the given answer in part (b) but many made fundamental errors in the laws of logarithms and gave an expression which was the sum of two terms. There was some evidence that candidates who had had $y = ax^b$ returned to their answer and changed it when they obtained $y = \ln a + b \ln x$ in part (b) which they did not recognise as being in the correct form. These two exemplars illustrate a successful and an unsuccessful attempt to use the given answer. Exemplar 4  	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	$\ln y = \ln(kx^n)$ $\ln y = \ln k + \ln(x^n) = \ln k + n \ln x$ SCHEME FOR CANDIDATES USING LOG BASE 10	M1 (AO2.1) E1 (AO2.1) [2] SC1	Taking natural logs of both sides and one correct use of laws of logs used. Convincing argument. AG Correct use of log instead of ln and no other error <u>Examiner's Comments</u> The method mark here was credited to candidates who took logarithms on both sides and demonstrated at least one correct use of the laws of logarithms, usually for correctly dealing with ln x^b or similar. Candidates who had the sum of terms in part (a) were not credited the mark for adding the logarithms of separate terms. Candidates who did not know how to answer part (a) could access many of the subsequent marks by working from the equation given in here in part (b). There was some evidence of candidates giving up on all of question 11 without realising that there were many marks accessible.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					Prepare candidates to work through questions working from a given result even if they are unable to show where that result came from. Exemplar 5  	
		c	<table><tr><th></th><th>ln$x$</th><th>ln$y$</th></tr><tr><td>Mercury</td><td>–1.179</td><td>9.575</td></tr><tr><td>Jupiter</td><td>1.599</td><td>4.022</td></tr></table> SCHEME FOR CANDIDATES USING LOG BASE 10 <table><tr><th></th><th>ln$x$</th><th>ln$y$</th></tr><tr><td>Mercury</td><td>–0.5122</td><td>4.158</td></tr><tr><td>Jupiter</td><td>0.6946</td><td>1.747</td></tr></table>	B1 (AO1.1b) B1 (AO1.1b) [2] B0 B1	At least 2 correct values All correct and to 4sf All correct. Must be 4 sf **Examiner's Comments** This was successfully done by the majority of candidates. It only required the use of the calculator and correct rounding to 4 significant figures – some candidates lost a mark for incorrect rounding. It was rare to see log10 used instead of natural logarithms.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	EITHER $b = \frac{9.575 - 4.022}{-1.179 - 1.599} = -1.999$ (-2.00 to 3sf) a = 7.218 OR 9.575 = a – 1.179b 4.022 = a + 1.599 b Giving a = 7.218 b = –1.999 (–2.00 to 3sf) SCHEME FOR CANDIDATES USING LOG BASE 10 As in main scheme a = 3.1336 b = –1.999	M1 (AO1.1a) A1 (AO3.1a) A1 (AO1.1b) [3] M1 A1 A1 [3] M1 A1 A1	using gradient formula Allow –2 a correct to at least 2 sf Setting up pair of equations by substitution of their values. Allow one slip. a correct to at least 2 sf b correct to at least 2 sf **Examiner's Comments** Candidates using the gradient method were often successful here. Many candidates using the alternative method were daunted by the complexity of the arithmetic here and made errors in the solution of their simultaneous equations.	These values could be found using the calculator STATS mode, so allow without working Simultaneous equations can be solved using calculator	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		e	$y = 1363x^{-2.00}$ SCHEME FOR CANDIDATES USING LOG BASE 10 $y = 1360x^{-2}$	B1 (AO2.2a) B1 (AO3.3) [2] B1 B1	FT their equation in (a) awrt 1300 or 1400, or $e^{7.2}$ or better. FT their *a* Allow for x^{-2} or better. FT their *b* FT their equation in (a) awrt 1300 or 1400, or $e^{3.1}$ (= 22.2), $10^{3.1}$ or better. FT their *a* allow for x^{-2} or better. FT their b **Examiner's Comments** Many candidates were hampered by incorrect work in parts (a) and (b) and so did not realise that e^a was required. It was possible to obtain both marks here using their values in the given equation in part (b) but this was rarely seen. Follow-through marks were given for using e^a and *b* in whatever equation had been seen in part (a).	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		f		B1 (AO1.2) B1 (AO1.1b) [2]	Appropriate curve with at least one horizontal asymptote or vertical asymptote shown Both asymptotes correct Ignore $x > 0$ if shown **Examiner's Comments** When attempted, many candidates correctly drew a graph with the positive x -axis as an asymptote but many thought their curve would cross the vertical axis at their a. As two points on the curve were given in the question, no marks were given for an increasing function.	FT their equation in (e) provided their function is a decreasing function	
		g	Earth $x = 1$, $y = 1363 \times 1^{-2}$ $= 1360$ W m^{-2} (3sf) SCHEME FOR CANDIDATES USING LOG BASE 10 Earth $x = 1$, $y = 1363 \times 1^{-2}$ $= 1360$ W m^{-2} (3sf)	B1 (AO3.4) [1] B1	FT their (e) FT their (e) **Examiner's Comments** This was an easy mark to obtain for candidates who substituted $x = 1$ into their equation even when the answer made little sense in the context.		
			Total	13			

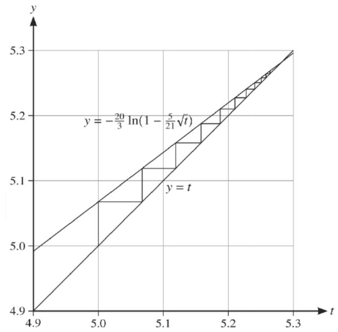
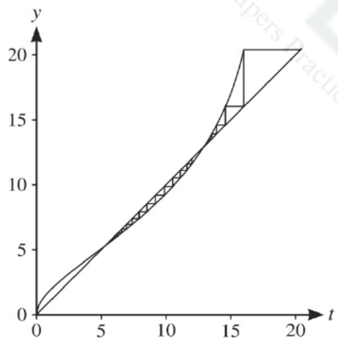
Question			Answer/Indicative content	Marks	Part marks and guidance		
44		a	$1 - 0.5^2$	B1 (AO 2.2a) [1]	oe		
		b	$\frac{0.5}{2}\{(2.718282+0.049787)+2(2.117+1+0.286505)\}$ = 2.39 (correct to 3 significant figures)	M1 (AO 1.1a) A1 (AO 1.1) [2]			
		c	$\frac{dy}{dx} = -2xe^{1-x^2}$ $\frac{dy}{dx} = 0$ only when $x = 0$, giving $y = e^{1-0} = e$	M1 (AO 1.1a) E1 (AO 2.1) [2]	 AG Convincing completion		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance	
45		a	$\log_{10}y = \log_{10}A + kx$ Equation is of straight line form '$Y = mX + c$', hence model is consistent with graph	M1 (AO 1.1) E1 (AO 2.4) [2]		
		b	$\log_{10}A = 4.83$ $A = 67600$ $k = \frac{4.83 - 4.6}{0 - 7}$ $k = -0.0328\ldots$	M1 (AO 1.1a) A1 (AO 2.2a) M1 (AO 1.1) A1 (AO 2.2a) [4]	Attempt to find gradient Allow –0.0325 to – 0.033	
		c	$\log_{10} 10000 < 4.83 - 0.0328x$ $x > 25.3$ 2036	M1 (AO 3.4) A1 (AO 1.1) A1 (AO 3.2a) [3]	Use of their equation	
		d	Not reliable, as extrapolation may not be valid	A1 (AO 3.5b) [3]	Accept any reasons for 'not reliable' that refer to possible future changes of circumstances, oe	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
46			5 $\log_a x$ or $\log_a x^{-1}$ oe soi	M1 (AO1.1)		
			6 $\log_a x$	A1 (AO1.1) [2]		
			Total	2		
47			$\int \frac{5}{y^2-y-6}\,\mathrm{d}y = \int \frac{1}{x}\,\mathrm{d}x$	M1 (AO 3.1a)	Attempt to separate variables	
			$\frac{5}{y^2-y-6} = \frac{A}{y-3} + \frac{B}{y+2}$	M1 (AO 3.1a)	Attempt to use partial fractions	With their factors
			$A = 1$, $B = -1$	A1 (AO 1.1b)	All correct	
			$\int \left(\frac{1}{y-3} - \frac{1}{y+2}\right) \mathrm{d}y = \int \frac{1}{x}\,\mathrm{d}x$	M1 (AO 1.1a)	Integrating, obtaining natural logs	
			$\Rightarrow \ln(y-3) - \ln(y+2) = \ln x + c$	A1 (AO 1.1b)	FT their partial fractions	
			When $x = 1$, $y = 8$ so $\ln 5 - \ln 10 = \ln 1 + c$	M1 (AO 1.1b)	Use of initial conditions to find c oe	
			$c = \ln\frac{1}{2}$	A1 (AO 1.1b)		
			$\ln(y-3) - \ln(y+2) = \ln x + \ln\frac{1}{2}$			
			So $\ln\left(\frac{y-3}{y+2}\right) = \ln\left(\frac{x}{2}\right) \Rightarrow \frac{y-3}{y+2} = \frac{x}{2}$	M1 (AO 3.1a)	Attempt to remove logs from their eqn	Of their equation
			$2y - = xy + 2x$ $y(2-x) = 6 + 2x$	M1 (AO 1.1a)	Attempt to make y the subject oe, but must be $y = f(x)$ form	
			$\Rightarrow y = \frac{6+2x}{2-x}$	A1 (AO 1.1b) [10]		
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
48		a	12000 = $a\sqrt{4}$ a = 5000 22000 –	B1 (AO 3.3) [1]	cao	
		b	$\frac{dV}{dt} = -5000 \times \frac{1}{2} \times t^{-\frac{1}{2}}$ $t = 4 \Rightarrow \frac{dV}{dt} = -1250$, so the caravan is losing value at a rate of £1250 per year when t = 4	M1 (AO 3.4) A1 (AO 1.1b) A1 (AO 3.4) [3]	Attempt to differentiate Correct derivative, in any form Must interpret the negative value of the derivative and must state units	
		c	For large values of t, [more than 19.36 years] the value of V is negative which is not possible	E1 (AO 3.5b) [1]	Any argument based on negative values for [sufficiently] large t	
		d	c = 1000 22 000 = be^0 + c so b = 21000	B1 (AO 3.3) B1 (AO 3.3) [2]	cao cao	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		e	Agree when $22\,000 - 5\,000\sqrt{t} = 21000e^{-0.15t} + 1000$ i.e. when $e^{-0.15t} = \frac{21\,000 - 5000\sqrt{t}}{21\,000} = 1 - \frac{5}{21}\sqrt{t}$ $-0.15t = \ln\left(1 - \frac{5}{21}\sqrt{t}\right)$ $t = -\frac{1}{0.15}\ln\left(1 - \frac{5}{21}\sqrt{t}\right)$ $= -\frac{20}{3}\ln\left(1 - \frac{5}{21}\sqrt{t}\right)$	M1 (AO 2.1) M1 (AO 2.1) A1 (AO 2.1) [3]	Equating their models and attempting to isolate the exponential term Taking logs of both sides AG; complete argument needed	$e^{-0.15t} = K$ need not be simplified at this stage	
		f	$t_1 = 5.067572851$ $t_2 = 5.118888519$	B1 (AO 1.1b) B1 (AO 1.1b) [2]	BC Allow answers which agree when rounded to 3sf		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		g		B1 (AO 1.1b) [1]	Must show at least two steps beginning at $t = 5$	
		h	The argument is false – the sequence is increasing and could [and in this case does] increase so that the root rounds to 5.3 or higher	B1 (AO 2.3) [1]	Accept an argument based on calculation of additional terms	Do not accept "false" without a reason
		i	 So the iteration will not find the root near 12 **Alternative solution 1** Sequence starting above the root, e.g.13, 13.035, 13.089, 13.174, 13.310, 13.532, 13.909 and	B1 (AO 2.4) B1 (AO 2.2a)	Sketch showing two staircases, one starting above the root and diverging and one starting below the root and converging on the root near 5	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			sequence starting below the root, e.g. 12, 11.611, 11.118, 10.530, 9.874, 9.193, 8.532, K	B1	Correct conclusion for correct reason		
			So the iteration will not find the root near 12	B1			
			Alternative solution 2				
			Near the root, the gradient of $y=-\frac{20}{3}\ln\left(1-\frac{5}{21}\sqrt{t}\right)$ is greater than 1	B1	Increasing sequence starting above and decreasing sequence starting below	Accept 'steeper than the line' for 'greater than 1'	
			So the iteration will not find the root near 12	B1	Correct conclusion for correct reason		
				[2]			
					For correct relevant statement about the gradient		
					Correct conclusion for correct reason		
			Total	16			

Question			Answer/Indicative content	Marks	Part marks and guidance	
49		a	ln a = b Area B: $\int_0^b e^y dy = e^b - e^0$ = a – 1 Area A: $\int_1^a \ln x \, dx = [x \ln x - x]_1^a$ = $a\ln a$ – a + 1 Their a – 1 = their $a\ln a$ – a + 1 $a\ln a$ – 2a + 2 = 0	B1 (AO 1.1) M1(AO 3.1a) A1(AO 1.1) M1(AO 2.1) A1(AO 1.1) M1(AO 3.1a) A1(AO 1.1) [7]	or a × b – their $a\ln a$ – a + 1	
		b	Evaluation of their f(4.921554 – δ) and their f(4.921554 + δ) eg – 0.000000079882 and 0.000000513742 seen to 2 or more sf plus correct conclusion: sign change, so Heidi is correct	M1 (AO 2.1) A1(AO 2.2a) [2]	$\delta \le 0.000\,000\,5$	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
50		a	5 million	B1(AO 3.4) [1]		
		b	The population is growing At a rate proportional to the population	B1(AO 3.2a) B1(AO 3.2a) [2]	**or** at 2% a year **or** by the same percentage each year	
		c	DR $5 \times 1.02^n = 10 \Rightarrow 1.02^n = 2$ $n = \frac{\log 2}{\log 1.02} = 35.002...$ 2035	M1(AO 3.4) M1(AO 1.1a) A1(AO 3.2a) [2]	oe	
		d	$\log_{10} P = \log_{10} 5 + n \log_{10} 1.02$ Of form $y = mx + c$ [with $\log_{10} P$ as y and n as x]	M1(AO 1.1) E1(AO 2.4) [2]	oe with clear link to gradient and intercept	
			Total	8		

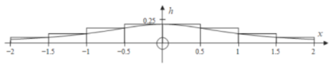
Question			Answer/Indicative content	Marks	Part marks and guidance	
51		a	DR $\frac{dy}{dx} = 1 + \ln x$ At stationary point ln x = –1 $x = \frac{1}{e}$ $y = -\frac{1}{e}$	B1(AO 1.1a) B1(AO 1.1) M1(AO 1.1) A1(AO 2.2a) A1(AO 2.2a) [5]	B1 for each term correct $\left(\frac{1}{e}, -\frac{1}{e}\right)$	
		b	DR $\frac{d^2y}{dx^2} = \frac{1}{x}$ At stationary point $\frac{d^2y}{dx^2} = e$ positive so minimum	M1(AO 1.1a) E1(AO 2.4) [2]	AG Convincing conclusion needed	
			Total	7		
52			e.g. 2 sticks of length $\sqrt{2}$ e.g. 4 sticks of length 1 $D = -\frac{\ln 2}{\ln\left(\frac{1}{\sqrt{2}}\right)} = 2$	M1 (AO 3.1a) M1(AO 2.1) A1(AO 1.1) [3]	Correct number of sticks for the chosen stick length Correct number of sticks for a different stick length AG Correct completion needed	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
53	a		**DR** $\ln 3^2$, $\ln 3^3$ seen $\ln 3 \times 2\ln 3 \times 3\ln 3 = 6(\ln 3)^3$	B1 (AO1.1a) B1 (AO2.2a) [2]	<u>Examiner's Comments</u> With this being a detailed reasoning question, it was expected that candidates show enough steps in their answers to provide detail of why each step follows. In this part, many went straight to $\ln 3 \times 2\ln 3 \times 3\ln 3$ without showing the intermediate step. This meant that those who got to $6(\ln 3)^3$ only scored 1 mark. Other fairly common mistakes were in misplacing the brackets in the final answer.	
	b		**DR** $3 > e$ so $\ln 3 > 1$ $(\ln 3)^3 > 1$ so $6(\ln 3)^3 > 6$	M1 (AO2.2a) E1 (AO2.4) [2]	Must mention e Convincing completion (answer given) <u>Examiner's Comments</u> Not very many candidates were successful in this part, with only a small proportion even using the fact that $e < 3$. Many tried finding numerical answers via their calculators rather than working with e.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
54	a		$m = \frac{3.58-3.93}{4-0}$ $\log_{10} V = -0.0875t + 3.93$	M1 (AO1.1) M1 (AO3.3) A1 (AO1.1) [3]	(±) 0.0875 oe Allow any value for – 0.0875 condone omission of base **Examiner's Comments** Only about half the candidates realised that they needed to find the gradient and intercept of the line in Fig. 9.2 and then form an equation for $\log_{10} V$ in terms of t. Some of these candidates made errors in finding the gradient of the line.	Condone ln
	b		$V = 10^{-0.0875t+3.93}$ $A = 10^{3.93} = 8500$ $b = 10^{-0.0875} = 0.82$	M1 (AO3.1b) A1 (AO3.3) A1 (AO1.1) [3]	Or $\log V = \log A + \log b^t$ Accept $V = 8500 \times 0.82^t$ **Examiner's Comments** Most candidates were able to find the value of A, but many had difficulty finding the value of b. Some candidates failed to give the answers correct to 2 significant figures.	8511.38 0.81752; sc B1 if both answers not to 2 sf. Allow 8500 taken from table (M1A1)

Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		3151.29 (˜ 3150) so the model is a (very) good fit	E1 (AO3.4) [1]	Must see value from model OR find value of t (approx. 4.98) when model gives 3150 **<u>Examiner's Comments</u>** This part was omitted by a considerable number of candidates. The command word 'determine' is intended to convey to candidates that a calculation may well be needed to justify their answer. Answers without both an approximately correct value and a correct conclusion were given no credit. Candidates were expected to show that the model in part (b) gives a value for the car of approximately £3151 after 5 years, and to conclude that the model was a (very) good fit. Answers were also accepted where candidates used more accurate values of A and b in the model, where candidates used the expression in part (a) to work out the value of the car after 5 years, and where candidates showed that the model would give the value of £3150 after about 4.98 years, which is very close to 5.	Eg 3107.93 from more accurate values.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
	d		$8500 \times 0.82^t = 500$ soi take logs of both sides ($t = 14.27$ so) after 15 years	M1 (AO3.1b) M1 (AO1.1) A1 (AO3.5a) [3]	Ft from (b) for M1M1 eg $\log 8500 + \log 0.82^t = \log 500$; can be implied by 15 or 14.(…) **Examiner's Comments** This part was often omitted. Most candidates tackled this using the model from part (b), though answers from the equation in part (a) were also accepted. Many of these arrived at a correct answer of (usually) 14.27 years, but then lost a mark by failing to round up to 15 years. A considerable number of candidates made errors in the manipulation of logarithmic expression.	or $\log 0.82^t = \log(\frac{500}{8500})$
			Total	10		

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th colspan="3">Part marks and guidance</th></tr>
<tr><td>55</td><td>a</td><td></td><td>
<table><tr><td>x</td><td>0</td><td>±0.5</td><td>±1</td><td>±1.5</td><td>2</td></tr><tr><td>h</td><td>$\frac{1}{5}$</td><td>$\frac{4}{25}$</td><td>$\frac{1}{10}$</td><td>$\frac{4}{65}$</td><td>$\frac{1}{25}$</td></tr><tr><td>h</td><td>0.2</td><td>0.16</td><td>0.1</td><td>0.061538</td><td>0.04</td></tr></table>
= 2 × 0.5(0.2 + 0.16 + 0.1 + 0.061538)

=0.522 to 3sf</td><td>M1(AO3.4)

M1(AO1.1a)
A1(AO1.1)
[3]</td><td>Finding at least 2 distinct values for h
Need not be a table of values

Using rectangles forming UB for area with width 0.5. Need not be drawn</td><td>Allow both M marks if only half the area considered.</td><td></td></tr>
<tr><td></td><td>b</td><td></td><td>Area $\approx \frac{0.5}{2}\left(0.2+0.04+2\left(0.16+0.1+0.061538\right)\right)$
=0.221 (to 3 sf)</td><td>M1(AO1.1a)
A1(1.1)
[2]</td><td>Using trapezium rule with 5 h values FT incorrect h values for the M mark
Allow awrt 0.22</td><td>Using the definite integral functionality of calculator gives 0.2214297 …
So method must be seen.</td><td></td></tr>
<tr><td></td><td>c</td><td></td><td>Volume = 2 × (0.2208 × 10)

= 4.42 m^3</td><td>M1(AO1.1a)

A1(AO1.1)
[2]</td><td>Condone missing factor of 2 for method mark
FT part (b)</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td>Total</td><td>7</td><td colspan="3"></td></tr>
</table>

Question			Answer/Indicative content	Marks	Part marks and guidance	
56	a		EITHER Take logs of both sides $x\log\left(\frac{1}{2}\right) < \log 4$ Giving $x > \frac{\log 4}{\log\left(\frac{1}{2}\right)}$ [since $\log\left(\frac{1}{2}\right)$ is negative] $x > -2$ OR Solve $\left(\frac{1}{2}\right)^x = 4$ by taking logs base $\frac{1}{2}$ $\log_{\frac{1}{2}}(4) = -2$ Test value – eg when $x = 0$ $\left(\frac{1}{2}\right)^0 = 1 < 4$ So $x > -2$	M1(AO2.1) B1(AO1.1a) A1(AO2.1) [3] M1 B1 A1 [3]	DR Use of laws of logs must be seen Allow equivalent with natural logs Award for the boundary value even if only evaluated. Correct inequality. DR Using log base $\frac{1}{2}$ Award for the boundary value even if only seen as part of an equation or incorrect inequality Correct inequality.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
	b		Using laws of logs $\log_2(x+8)^2 - \log_2(x+6) = 3$ $\log_2 \frac{(x+8)^2}{(x+6)} = 3$ $\frac{(x+8)^2}{(x+6)} = 2^3$ $(x+8)^2 = 8(x+6)$ $x^2 + 8x + 16 = 0$ Discriminant is $8^2 - 4 \times 1 \times 16 = 0$ So there is only one solution	M1(AO3.1a) M1(AO3.1a) A1(AO1.1) M1(AO2.1) A1(AO2.2a) [5]	**DR** At least one correct use of laws of logs Clearing logs to obtain 2^3 or 8 seen in an equation Correct quadratic Attempt to find the discriminant of their quadratic (allow one slip) Correct argument from zero discriminant or repeated root found $x = -4$	Allow M1 for an attempt to solve their quadratic	
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance	
57	a	i	ke^{kx}	B1 (AO1.2) [1]	**Examiner's Comments** This question says 'Show that the straight line is consistent …' so an answer should make clear how any working has shown the consistency and conclude that 'This shows the straight line is consistent with this model'.	
		ii	Reason referring to growth proportional to profit.	E1 (AO3.3) [1]	E.g. As more people hear about the business, they will sell more so it is reasonable that the rate of growth is proportional to the profits	
	b		$ln\ y = lnA + kx$ Equation is of form "$y = mx + c$" so a straight line - hence model is consistent with graph	M1 (AO1.1) E1 (AO2.4) [2]		

Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		*ln A* = 1.9 *A* = 6.686 $k = \frac{0.4}{1.6}$ *k* = 0.24 to 0.25	M1 (AO1.1a) A1 (AO2.2a) M1 (AO1.1) A1 (AO2.2a) [4]	Intercept = 1.9 not enough for M1 One or more d.p. Attempt to find gradient	Allow $e^{1.9}$ $\frac{11}{45}$ gets M1A1	
	d		$x = 11$ $y = 6.686 \times e^{0.25x}$ 104.586… £104587	B1 (AO3.3) M1 (AO3.4) A1 (AO3.2a) [3]	Use of model with *their A* and *k* Translation into pounds (may be to nearest thousand) (FT their *k* and *a*)		

Question			Answer/Indicative content	Marks	Part marks and guidance	
	e		Not reliable. Extrapolation may not be valid o.e. OR Fit of model is good so far and it's only two more years o.e.	E1 (AO3.5b) [1]	E.g. • The relationship between sales and time may change • There is a limit to the market for revision resources • Changes in the economy may affect the business	'Reliable' plus 'extrapolation' does not score
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance		
58	a		$\frac{dy}{dx}=\frac{\frac{x}{x}-\ln x}{x^2}$ $\frac{1-\ln x}{x^2}=0 \Rightarrow \ln x=1 \Rightarrow \quad x=e$	M1 (AO1.1a) A1 (AO1.1) E1 (AO2.2a) [3]	M1 for attempt to use quotient rule (allow one error) Convincing completion (AG)	Subbing x = e gets M0	
	b		$$\frac{dy}{dx}=\frac{1}{x^2}-\frac{\ln x}{x^2}$$ $\frac{d^2y}{dx^2}=-\frac{2}{x^3}-\frac{\frac{x^2}{x}-2x\ln x}{x^4}$ $=-\frac{2}{x^3}-\frac{x-2x\ln x}{x^4}$ or $\frac{x^2\left(-\frac{1}{x}\right)-(1-\ln x)2x}{x^4}$ $=\frac{-3+2\ln x}{x^3}$ When $x=e, \frac{d^2y}{dx^2}=-\frac{1}{e^3}<0$ hence maximum	M1 (AO1.1a) A1 (AO1.1) E1 (AO2.1) [3]	Attempt to differentiate again (allow one error) Correct second derivative Correct conclusion from correct working	OR M1 subst (must be seen) values either side of e into derivative A1 correct conclusion about sign of gradient E1 correct conclusion from correct working regarding maximum (-0.05)	
	c		$\frac{\ln e}{e}>\frac{\ln a}{a}\Rightarrow\frac{1}{e}>\frac{\ln a}{a}$ or $a\ln e>e\ln a$ $e^{\frac{a}{e}}>a$ hence $e^a>a^e$ or $\ln e^a>\ln a^e$ $e^a>a^e$	M1 (AO3.1a) A1 (AO2.4) [2]	Convincing completion (AG)		
			Total	8			

Question			Answer/Indicative content	Marks	Part marks and guidance	
59	a		$\frac{A}{x(A-x)}$	B1 (AO1.1) [1]	Denominator may be multiplied out **Examiner's Comments** Only the strongest candidates appreciated the connection between this question parts (a) and (b) and many attempted, unnecessarily, a partial fraction decomposition with variable degrees of success. Not all candidates appreciated the need to give the number of fish in the lake as an integer. AfL The front cover instructions state that candidates should 'Give your final answer to a degree of accuracy that is appropriate to the context'. This may will be 3 significant figures in most cases, but it is always good practice to check back with the question before stating the final numerical answer.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
	b		DR $\int \frac{400}{x(400-x)}\,dx = \int dt$ $\int \left(\frac{1}{x} + \frac{1}{400-x}\right) dx = \int dt$ $ln\ x - ln(400 - x) = t + c$ $ln\left(\frac{x}{400-x}\right) = t + c$ When $t = 0$, $x = 100$ so $c = ln\left(\frac{1}{3}\right)$ $t = ln\left(\frac{3x}{400-x}\right)$ When $t = 10$, $10 = ln\left(\frac{3x}{400-x}\right)$ $\frac{3x}{400-x} = e^{10} \Rightarrow 3x = e^{10}(400-x)$ 400 *OR* DR	M1 (AO1.1a) M1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) M1 (AO1.1) M1 (AO3.4) M1 (AO2.1) A1 (AO3.2a) M1 M1 M1 A1 M1 M1 M1 A1 [8]	Separates variables (inc dx and dt) Use of result from **(a)** or method for partial fractions Attempt to integrate (at least one term correct) All integration correct Or $ln\left\|\frac{Ax}{400-x}\right\| = t$ Finding constant (or A = 3) Using log laws to combine the c value Attempt to rearrange to find x Must be rounded to nearest whole number Separates variables Use result from (a) or method for partial fractions Attempt to	400 either side	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			$\int_{100}^{X} \frac{400}{x(400-x)} dx = \int_0^{10} dt$ $\int_{100}^{X} \left(\frac{1}{x} + \frac{1}{400-x}\right) dx = \int_0^{10} dt$ $[\ln x - \ln(400-x)]_{100}^{X} = [t]_0^{10}$ $\left[\ln \frac{x}{(400-x)}\right]_{100}^{X} = [t]_0^{10}$ $\ln \frac{X}{(400-X)} - \ln \frac{100}{300} = 10 - 0$ $\ln \frac{X}{(400-X)} - \ln \frac{1}{3} = 10$ $\ln \frac{3X}{(400-X)} = 10$ $\frac{3X}{(400-X)} = e^{10}$ $3X = e^{10}(400-X)$ $X = 400$		integrate (at least one term correct) All integration correct Limits applied (condone one error) Combining log terms together Attempt to rearrange to find X Must be rounded to nearest whole number		
			Total	9			

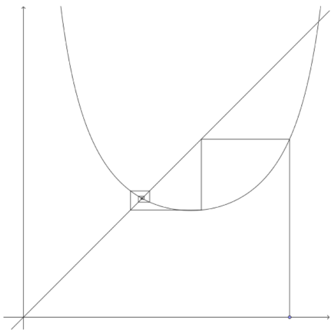
Question			Answer/Indicative content	Marks	Part marks and guidance	
60	a		$A = 11$ Substitution of $t = 3$ and $V = 13.8$ $B = 7.0$	B1 (AO1.1) M1 (AO3.3) A1 (AO1.1) [3]	from $t = 0$ and $V = 11$ $13.8 = A + B(1 - e^{-0.17\times3})$ Allow 7, 7.01, 7.009, etc	Implied by 7.0086
					<u>Examiner's Comments</u> This question was done extremely badly because very few candidates realised that the speed of the car is 11 m s^{-1} when the car starts to accelerate, which is when $t = 0$. As a result, few candidates found the value of A to be 11, though other candidates gained partial credit. In part (c) very few candidates used differentiation to find acceleration, though equations for motion with constant acceleration were seen quite often. While only a small number of candidates succeeded in part (d), some gained partial credit for converting a speed from m s^{-1} to km h^{-1} or vice versa. **Misconception** Candidates did not understand that points on a cumulative frequency diagram are plotted at the upper end of each interval in Question 2(a). They often confused 'Quota sampling' with 'Systematic sampling' in Question 3(a) and were unable to explain how to carry out systematic sampling in Question 10(f). Many candidates did not make progress on Question	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					11 because they did not realise that $e^0 = 1$.	
	b		$t = 4$ and $V = 14.453681053\ldots$to 1 or more dp good fit $t = 5$ and $V = 15.008095476..$to 1 or more dp good fit	B1 (AO3.4) B1 (AO1.1) [2]	allow **SC1** for two correct values with no comment or incorrect comment(s) If B0B0, allow sc M1 for subst with their values.	Allow full marks if more accurate value of *B* used – 14.458 and 15.013
	c		$\frac{dV}{dt} =$ their $7.0 \times 0.17e^{-0.85}$ 0.509 ms^{-2}	M1 (AO31.a) A1 (AO1.1) [2]	Allow for their $7.0 \times 0.17\, e^{-0.17t}$; condone use of 7.0086 for *B*	
	d		$t \to \infty$, $V \to 11 + 7$ oe their 18×3.6 $64.8 > 60$ so the motorist is fined	M1 (AO3.3) M1 (AO1.1) A1 (AO3.5a) [3]	Allow ft from (a) for M1M1 Allow for any attempt to change to km / hr, accept use of 'their 18' Need to show 64.8 > 60 for full marks	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance		
61			1 – cosec1 = –0.188… or ‘negative’ 2 – cosec 2 = 0.900 or ‘positive’ Change of sign so root between 1 and 2	B1 (AO1.1a) E1 (AO2.4) [2]	Both correct Condone no mention of continuity AG	OE E,g, may use cosec$x$ – x Dep on B mark	
			Total	2			

Question			Answer/Indicative content	Marks	Part marks and guidance	
62	a		$\sin 2x \approx 2x$ or $\sin x \approx x$ used $\int \left(\frac{1}{x}\right) \mathrm{d}x$ or $\int \left(\frac{1}{x} - x\right) \mathrm{d}x$ obtained **oe nfww** F[x] = ln$x$ **oe** or F[x] = ln$x$ – ½ x^2**oe** ln(0.05) – ln(0.01) = ln5 **oe** or ln(0.05) – ln(0.01) + 0.0012 ≈ ln5 **oe**	M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) A1 (AO3.2a) [4]	$\cos x \approx 1 - \frac{x^2}{2}$ may see intermediate step needed from here to earn final mark	
	b		differentiation of *their* $\frac{1}{x}$ substitution of 0.01 and –10 000 correctly obtained	M1 (AO2.1) A1 (AO1.1) [2]	or differentiation of y using quotient rule and use of small angle approximation from $-\frac{1}{x^2}$ or $-\frac{1}{x^2} - 1$ **oe**	
	c		4.54066×10^{-6} or 0.00000454066 **cao** (no sign change for 6 dp), but sign change for 5 dp or last two iterates agree to 5dp	B1 (AO2.5) E1 (AO3.1a) B1 (AO3.2a)	allow sign change between 0.947745 and 0.9477475 <u>**Examiner's Comments**</u>	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			0.94775	[3]	Candidates who did well in this question made efficient use of the small angle approximation for sin x and sensibly used $\sin 2x \approx 2x$ to obtain $\int \frac{1}{x}\,dx$, or used the double angle formula and then the small angle approximation for x. Full marks in parts (a) and (b) was commonly achieved. Those who did less well did not simplify their expression for $\frac{1}{x}$ fully before integrating and went on to make a slip (often integrating by parts) and similarly used the product or quotient rule in part (b) to differentiate $\frac{x}{x^2}$ for example. Candidates who did well in part (c) were clearly familiar with spreadsheets and were able to give a clear and correct response to both requests. Candidates who did less well either didn't understand the spreadsheet notation, or didn't appreciate the significance of the change of sign in the table.	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance		
63	a			B1 (AO3.2a) B1 (AO2.2a) B1 (AO1.1) [3]	Starting point between min and right asymptote Initial “staircase” (≥ 2 horiz sections) Spirals into lower root	3 B marks all indepen dent)	
	b		No, it converges to 1.114	E1 (AO1.1) [1]	OR same as *their* **(c) (ii)** or ‘the root between 1 and 2’ etc	Just ‘No’ gets 0 ‘Yes’ with anything gets 0	
	c	i	**BC** 1.18840…, 1.07785…	B1 (AO1.1a) [1]	Both correct to at least 3dp.		
		ii	**BC** 1.114	B1 (AO2.2a) [1]			
			Total	6			